PHYSICS GRADE 10: ENERGY, WORK, POWER AND MACHINES

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.4 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 1.0: Mechanics and Thermal Physics
Sub-Strand	Sub-Strand 1.6: Energy, Work, Power and Machines
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Forms of energy: kinetic energy and gravitational potential energy (definitions, formulae, and calculations) – Conservation of mechanical energy: transformation between kinetic and potential energy in closed systems – Work: definition, formula ($W = Fd\cos\theta$), positive work, negative work, and zero work – Work done against friction: energy losses and heat generation in real-world systems – Power: definition, formula ($P = W/t$), units (watts, kilowatts), and everyday applications – Simple machines: types (lever, pulley, inclined plane, wheel and axle, screw, wedge) and their functions – Mechanical advantage (MA), velocity ratio (VR), and efficiency of machines – Applications of simple machines in Kenyan agriculture, construction, and transport contexts
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) define kinetic energy and gravitational potential energy and apply their formulae to solve numerical problems, b) explain the principle of conservation of mechanical energy and demonstrate energy transformations between kinetic and potential forms, c) define work, state its SI unit, and calculate work done including cases of positive, negative, and zero work, d) calculate work done against friction and explain how friction leads to energy dissipation as heat, e) define power, state its SI unit, and solve problems involving power in everyday contexts such as Kenyan athletes running and water pumping, f) identify and describe the types of simple machines and explain how they make work easier, g) calculate the mechanical advantage, velocity ratio, and efficiency of simple machines, h) appreciate the role of energy, work, power, and machines in solving real-world problems in Kenyan society.
Core Competencies	– Communication and Collaboration: Learners work in groups to investigate simple machines (e.g., constructing a model lever or pulley using locally available materials) and present their findings on mechanical advantage and efficiency to the class, embracing teamwork and shared decision-making. – Critical Thinking and Problem Solving: Learners analyse energy transformation scenarios (e.g., a maize grain falling from a grain store in Kitale) and solve multi-step numerical problems involving work, power, KE, PE, and machine efficiency. – Learning to Learn: Learners use PhET simulations (Energy Skate Park, Pendulum Lab) and Khan Academy interactive exercises to independently consolidate understanding of energy conservation and power, practising self-directed knowledge construction.

	– Digital Literacy: Learners interact with PhET simulations, LabXchange modules, and YouTube videos on energy transformations and simple machines, developing competence in using digital tools for science learning. – Creativity and Imagination: Learners design and build simple machines from locally sourced materials (e.g., a wheelbarrow model or an inclined plane using planks) to creatively solve a given load-lifting challenge in the school environment.
Core Values	– Responsibility: Learners take ownership of their roles during group investigations of simple machines and energy experiments, ensuring accurate data collection and honest reporting of results. – Respect: Learners demonstrate open-mindedness and appreciation for diverse approaches when peer groups present different machine designs and efficiency calculations. – Unity: Learners collaborate across different backgrounds and abilities during hands-on machine-building activities, recognising that collective effort produces better outcomes — mirroring how community members in Kenyan villages cooperate to use pulleys and levers on construction projects. – Social Justice: Learners examine how access to simple machines and energy-saving technologies can reduce the physical burden on women and children who carry heavy loads in rural Kenya, connecting physics to equity concerns.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners generate investigable questions from the anchoring phenomenon (e.g., "Why does the jua kali mechanic use a long-handled spanner rather than a short one?") and refine them into testable problems about force, distance, and work. – Planning and Carrying Out Investigations: Learners design and conduct experiments to measure the mechanical advantage and efficiency of a pulley system or inclined plane using spring balances, rulers, and locally sourced loads (e.g., bags of sand or stones). – Analysing and Interpreting Data: Learners record force and distance data from machine experiments, calculate MA, VR, and efficiency, plot graphs (e.g., load vs. effort), and identify patterns and sources of energy loss. – Using Mathematics and Computational Thinking: Learners apply formulae for KE, PE, W, P, MA, VR, and efficiency in structured and open-ended numerical problems set in Kenyan contexts (e.g., a farmer in Kericho lifting tea bales using a pulley). – Constructing Explanations: Learners construct written and oral explanations of why machines cannot be 100% efficient by referencing friction and energy dissipation observed in their investigations. – Obtaining, Evaluating, and Communicating Information: Learners evaluate video demonstrations (YouTube), PhET simulations, and LabXchange modules, then communicate their understanding through model diagrams, summary tables, and class presentations.
Pertinent & Contemporary Issues (PCIs)	– Life Skills Education (LSE) — Health and Physical Activity: Learners connect the concept of power to human physical exertion, exploring how Kenyan athletes such as marathon runners (e.g., Eliud Kipchoge) generate mechanical power and how energy efficiency relates to physical fitness and healthy lifestyles. – Environmental Education: Learners examine how energy losses due to friction and heat generation contribute to inefficiency, linking this to the importance of energy conservation in Kenya's context of rising electricity costs and the push for renewable energy sources. – Financial Literacy: Learners calculate the cost implications of using machines of different efficiencies (e.g., comparing a manual water pump to an electric pump in a rural setting), developing awareness of economic decision-making linked to energy and power use. – Gender Disparity: Learners discuss how simple machines can reduce the disproportionate physical burden borne by women and girls in Kenya (e.g., fetching water, grinding grain), exploring how technology and physics knowledge can promote gender equity in labour.
Career Connections	– Mechanical Engineering: This sub-strand directly underpins the design and analysis of machines, engines, and mechanical systems used in Kenya's manufacturing and construction industries.

	– Agricultural Engineering: Understanding simple machines, efficiency, and power is essential for designing and operating farm machinery such as ox-ploughs, grain mills, and irrigation pumps used across Kenya's Rift Valley and Central regions. – Civil Engineering and Construction: Work, force, and machine principles apply directly to the use of cranes, pulleys, inclined planes, and levers on Kenyan construction sites. – Sports Science and Physiotherapy: The concepts of kinetic energy, power, and work relate to biomechanics — informing training programmes for Kenyan athletes at institutions such as the High Altitude Training Centre in Iten. – Renewable Energy Technology: Understanding energy transformation and power calculations is foundational for careers in solar, wind, and hydroelectric energy sectors — all of which are growing rapidly in Kenya. – Physics Teaching: Mastery of core mechanics concepts equips learners to become physics educators, addressing the shortage of qualified science teachers in Kenyan secondary schools.
Focus for Lessons	This unit develops learners' understanding of mechanical energy (kinetic and potential), the principle of conservation of energy, and the concepts of work and power — building towards an analysis of how simple machines exploit these principles to make tasks easier in everyday Kenyan life. Learners engage with phenomena drawn from familiar local contexts — jua kali workshops, tea farms in Kericho, and athletes in Iten — to ground abstract physics principles in tangible, meaningful experiences. The pedagogical approach combines PhET simulations, LabXchange modules, Khan Academy practice, and hands-on machine investigations using locally available materials, culminating in a cumulative model that explains why no machine is perfectly efficient.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why can a jua kali mechanic in Nairobi loosen a wheel nut effortlessly with a long spanner when a short one fails — and where does the energy go when friction makes the nut hot? KICD KEY INQUIRY QUESTIONS: 1. How is energy conserved and transformed in mechanical systems? 2. How do simple machines make work easier without creating extra energy?
Anchoring Phenomenon	A jua kali (informal sector) mechanic in Gikomba market, Nairobi, is removing a stubborn wheel nut from a matatu using an extra-long spanner handle extended with a metal pipe. Despite pushing hard with a short spanner and failing, he succeeds effortlessly with the long extension. Meanwhile, a nearby apprentice notices that the wheel nut feels warm after several failed attempts with the short spanner. Students observe that: (1) the same nut is loosened with far less muscular effort when the spanner arm is made longer; and (2) energy seems to "disappear" as heat during the struggle. This is puzzling because: if the machine (the spanner-lever system) makes the job easier, where does the extra force come from? And if energy is conserved, why does the nut get hot and why can't we get more work out than we put in? These two intertwined mysteries — mechanical advantage and energy dissipation — anchor the entire unit's investigation into energy, work, power, and machines.
Storyline Thread	Lesson 1 — Kinetic Energy and Gravitational Potential Energy: Learners observe a video of a Kenyan athlete sprinting and a stone falling from a grain store roof in Kitale. They predict which has more energy and why, then derive $KE = \frac{1}{2}mv^2$ and $PE = mgh$ through guided inquiry using the PhET Energy Skate Park simulation. Students add their first "wonder" questions to the Driving Question Board (DQB). Lesson 2 — Conservation of Mechanical Energy: Using the PhET Pendulum Lab and a YouTube pendulum-and-roller-coaster demonstration, learners investigate how KE and PE continuously interchange. They observe that the total mechanical energy remains constant in a frictionless system, building a class energy transformation model diagram. The DQB is updated: "But if energy is conserved, why does the wheel nut get hot?" Lesson 3 — Work: Definition and Calculation: Learners are formally introduced to $W = Fd \cos\theta$. Through a structured activity (pushing a school desk across the classroom floor), they distinguish positive work (push in direction of motion), negative work

	(friction opposing motion), and zero work (carrying a bag horizontally). The jua kali spanner scenario is revisited: how much work does the mechanic do with the short vs. long handle? Lesson 4 — Work Done Against Friction: Learners conduct an investigation using a wooden block, a spring balance, and surfaces of different roughness (smooth tile, sandpaper) to measure the force needed to overcome friction and calculate work done against friction. They explain why the wheel nut heats up: work done against friction converts mechanical energy to thermal energy. The class model is updated to include an energy dissipation "leak" arrow. Lesson 5 — Power: Definition, Calculation, and Everyday Applications: Learners watch a YouTube stair-climbing demo and calculate the power output of a student vs. a teacher climbing the same staircase at different speeds. They connect this to Kipchoge's marathon power output and to the wattage rating of the Mwea irrigation pump. Khan Academy interactive problems consolidate calculations. The DQB is revisited: "So a more powerful machine does the same work faster — but is it more efficient?" Lesson 6 — Simple Machines: Types and Mechanical Advantage: Learners identify six types of simple machines in photographs of Kenyan contexts (a jua kali workshop lever, a Kisumu harbour pulley, a Kericho tea farm inclined ramp, a borehole hand-pump wheel and axle, a screw press for sugarcane, and a wedge-shaped panga/machete). They measure MA = Load ÷ Effort using spring balances and constructed models, linking back to why the long spanner handle works. Lesson 7 — Velocity Ratio and Efficiency: Learners derive VR from distance measurements on their machine models and calculate efficiency = (MA/VR) × 100%. They discover no machine achieves 100% efficiency and connect this to their earlier finding about friction converting work to heat. The cumulative class model is updated to show that machines redistribute force and distance but cannot create energy — and friction always "steals" some. Lesson 8 — Synthesis, Model Finalisation, and DQB Resolution: Learners revisit the anchoring phenomenon (the jua kali mechanic) and use their full conceptual model to explain: (1) why the long spanner creates mechanical advantage through leverage, (2) why the nut gets hot due to work done against friction, and (3) why the mechanic cannot get more work out than he puts in. Groups present their final annotated models. The DQB is reviewed and all original questions are answered or redirected to future learning. A summary table of energy, work, power, and machine formulae is co-constructed with the class.
Supporting Phenomena	– A Kericho tea farmer uses a simple rope-and-pulley system to hoist heavy sacks of picked tea leaves from the base of a hillside to a collection point above — using less force but pulling a much longer length of rope. Why does this "feel" easier even though the sack weighs the same? – Eliud Kipchoge running the Nairobi Marathon sustains an extraordinary rate of energy output (power) for over two hours. A physics student climbing the same stairs as Kipchoge at a slower pace does the same total work but takes much longer — what does this reveal about the difference between work and power? – A water pump on a smallholder farm in Mwea Irrigation Scheme is rated at 500 W but only lifts water at 60% of its theoretical maximum rate. Where does the "missing" 40% of energy go, and how does this connect to what the mechanic's warm wheel nut tells us about friction? – A child in Kisumu rolls a metal barrel up an inclined ramp onto a truck bed rather than lifting it straight up. The child exerts less force over a longer distance. How does the inclined plane "trade" force for distance, and is any energy actually saved?

LESSON 1 (80 minutes): Kinetic Energy and Gravitational Potential Energy: What Makes Things Move and Fall?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce learners to kinetic energy (KE = $\frac{1}{2}mv^2$) and gravitational potential energy (PE = mgh) through guided inquiry, anchored in the jua kali mechanic phenomenon, so that learners can begin to explain where energy comes from and where it goes in mechanical systems.
Knowledge	By the end of this lesson, the learner should be able to: (a) define kinetic energy as the energy an object possesses due to its motion; (b) define gravitational potential energy as the energy stored in an object due to its position above a reference point; (c) state and apply the equations KE = $\frac{1}{2}mv^2$ and PE = mgh to solve problems in context.
Skills	By the end of this lesson, the learner should be able to: (a) predict and compare energy values of moving and elevated objects using prior knowledge; (b) gather evidence from a simulation (PhET Energy Skate Park) to derive the mathematical relationships for KE and PE; (c) construct initial scientific explanations linking observed energy changes to the anchoring phenomenon.
Attitudes	By the end of this lesson, the learner should be able to: (a) appreciate that curiosity and careful observation are the starting points of scientific inquiry; (b) demonstrate open-mindedness by revising initial predictions when evidence contradicts them; (c) value the relevance of physics in familiar Kenyan everyday contexts such as the jua kali economy.
Key Inquiry Question	How is energy conserved and transformed in mechanical systems?
Purpose in Storyline	This is the opening lesson of the unit. Learners encounter the anchoring phenomenon — a jua kali mechanic in Gikomba, Nairobi, struggling with a short spanner and succeeding with a long one, while a wheel nut grows hot from friction — and immediately face two puzzles: where does the extra force come from, and where does the energy 'disappear' to? Before those puzzles can be resolved, learners must first build a precise understanding of what energy actually is and how to measure it. This lesson establishes KE and PE as the foundational energy forms, giving students the first conceptual tools they will use across all subsequent lessons to trace energy through mechanical systems, simple machines, work, and power.
Safety Notes	No hazardous materials are used in this lesson. If learners physically demonstrate throwing or dropping objects during the Predict phase, ensure the classroom space is clear of obstacles and that objects used are soft (e.g., a rolled-up exercise book). Devices used for the PhET simulation must be handled responsibly. Learners must not stand on chairs or desks at any point during demonstrations.

B. LESSON OVERVIEW

Lesson 1 launches the unit by immersing learners in the anchoring phenomenon: a video/image vignette of a jua kali mechanic at Gikomba market using an extended spanner on a matatu wheel nut, and a stone falling from a grain-store roof in Kitale. Learners first predict which of two objects — a sprinting Kenyan athlete or a falling stone — carries more energy, and justify their reasoning in writing. They then gather evidence using the PhET Energy Skate Park simulation and a guided data table to explore how mass and velocity affect KE, and how height affects PE. Through a class Claim–Evidence–Reasoning (CER) discussion, learners derive and make sense of KE = $\frac{1}{2}mv^2$ and PE = mgh. The lesson closes with learners posting their first "wonder" questions onto the class Driving Question Board (DQB), officially opening the unit-long investigation. All phases are explicitly tied back to the mechanic phenomenon: the spanner arm moving through an arc stores and releases energy, and the heat felt in the wheel nut is energy that has gone somewhere — but where?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Conservation of energy] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [SI Kinetic Energy Units (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners watch a 60-second teacher-narrated slide show (or printed image sheet if devices are unavailable) showing: (1) a Kenyan sprinter (e.g., Ferdinand Omanyala) running at full speed on a track; (2) a large stone tumbling from the roof of a grain store in Kitale. Learners are then given a silent 3-minute individual writing task: 'Which object has more energy — the sprinting athlete or the falling stone? Write two reasons for your choice. What do you think energy actually IS?' They then share with a partner (Think-Pair-Share, 2 minutes) before three cold-call responses are taken by the teacher. Learners also receive their first look at the anchoring phenomenon: a photograph of the jua kali mechanic at Gikomba using the long-pipe spanner extension. Teacher asks: 'The mechanic pushes the long spanner and the nut moves. The apprentice notices the nut is warm. Where is the energy in this picture?' Learners write one sentence prediction in their exercise books.	1. Display the sprinter and falling-stone images side by side. Say exactly: 'Look at these two objects. Both are moving. In your exercise book, write your answer to this question: Which one has more energy, and what two reasons support your choice? You have 3 minutes — work in silence.' START a visible 3-minute timer. 2. After 3 minutes, say: 'Now turn to your partner. You have 2 minutes — share your answer and listen carefully to theirs. Do you agree or disagree?' 3. After 2 minutes, cold-call EXACTLY 3 learners (choose one who appears confident, one who appears unsure, one who has not yet spoken). Ask each: 'What did you and your partner decide, and what was your reasoning?' Apply 10-second WAIT TIME after each question before accepting a response. Record key words from responses on the board in two columns: 'Speed matters' and 'Size/mass matters'. 4. Display the Gikomba mechanic photograph. Say: 'Hold that thought. Here is the mystery we are going to investigate for the next several weeks. A mechanic uses a longer spanner and suddenly the job is easy. The nut gets hot. Where is the energy in this picture?' Give 2 minutes silent writing. Do NOT correct or affirm predictions — say: 'We are not answering this today. We are collecting our first guesses. Be honest about what you don't know.' 5. Affirm intellectual risk-taking: 'There are no wrong predictions in science — only predictions that evidence will help us improve.'	Think-Pair-Share with Public Record: By making learners commit to a written prediction before discussion, the teacher surfaces prior conceptions (and misconceptions) about energy as 'strength,' 'speed,' or 'size.' Recording key words publicly on the board creates a shared reference that the class will return to at the end of the lesson to see which ideas were confirmed or revised. This strategy builds metacognitive awareness — learners begin to see their own thinking as something that changes with evidence.	OBSERVABLE INDICATOR: Each learner has a written prediction in their exercise book with at least two reasons before pair-share begins. Teacher circulates during silent writing and notes (without correcting) whether learners mention mass, speed, height, or unrelated ideas (e.g., 'energy = strength'). Cold-call responses reveal whether learners can distinguish between force and energy. RED FLAG: If fewer than half the class can name even one factor affecting energy, the teacher spends an additional 2 minutes eliciting everyday examples ('When does ugali give you the most energy — after eating a little or a lot?') to bridge from nutritional energy to mechanical energy before proceeding.

Observe Phase 📹 **VIDEO:** Conservation of energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** SI Kinetic Energy Units (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open the PhET Energy Skate Park simulation (online or pre-loaded offline on ARES devices). Working in pairs, they complete a structured data-gathering activity in two parts. PART A — Kinetic Energy: Learners keep the skater's height at zero (flat surface) and vary mass (30 kg, 60 kg, 90 kg) and speed (1 m/s, 2 m/s, 3 m/s) using the simulation sliders. They record KE values shown by the simulation's energy bar into a provided data table (6 rows). PART B — Gravitational PE: Learners keep mass constant at 60 kg and vary the starting height (1 m, 2 m, 3 m) on the skate ramp. They record PE values. Between Part A and Part B, the teacher plays a 2-minute YouTube clip (pre-downloaded) showing a stone being dropped from different heights at a construction site in Kisumu — learners observe that it makes a bigger dent in the sand the higher it is dropped from. Learners annotate their data tables with a 'pattern sentence': 'As [variable] increases, KE/PE ___________.'	1. Before opening devices, distribute the printed data table. Say: 'You are scientists today. Your job is NOT to play the simulation — your job is to gather evidence. Every number you record is a piece of evidence. Let's be precise.' 2. Demonstrate the simulation for 90 seconds on the projector: show how to change mass, speed, and height sliders. Say: 'Read the KE bar — the number appears here. Record it exactly. Do not round.' 3. Set a 12-minute timer for data collection. Circulate and ask targeted questions to pairs who finish early: 'What happens to KE when you DOUBLE the speed? Does it double too? Check your data.' Apply 15-second WAIT TIME before offering any hint. 4. After Part A data is collected (approximately 7 minutes), pause the class. Say: 'Before we look at height, I want to show you something from a building site in Kisumu.' Play the 2-minute video. Ask immediately after: 'What variable changed in that video? What was the effect?' Cold-call 2 learners. 5. Release learners to Part B (5 minutes). 6. After data collection ends, say: 'Write ONE sentence in your data table that describes the pattern you see. Start with the words: As height increases…' Give 2 minutes. Cold-call 3 learners to read their sentences aloud. Write the three sentences verbatim on the board without editing. 7. Bridge back to phenomenon: 'The mechanic's spanner arm moves through an arc — it has height AND speed at different points. Keep that image in your mind.'	Data-Table Pattern Recognition: Learners are not told the equations first — they are led to notice that doubling speed quadruples KE (from their data), while doubling mass only doubles KE. This productive struggle with numbers primes learners to find the equations meaningful rather than arbitrary. The Kisumu video clip provides a concrete real-world anchor for PE before the abstract equation is introduced, grounding the mathematics in observable reality.	OBSERVABLE INDICATOR: Each pair has a completed data table with 6 KE entries and 3 PE entries, plus a written pattern sentence for each. Teacher checks that at least one pair per group has noticed the quadrupling relationship for speed (KE quadruples when speed doubles). If no pair notices this, teacher intervenes: 'Look at your KE when speed = 1 m/s and KE when speed = 2 m/s. What is the ratio? Now look at the speeds — what is that ratio?' This targeted question, without giving the answer, redirects learners to the key evidence without abandoning the inquiry approach.

Explain Phase 📹 **VIDEO:** Conservation of energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** SI Kinetic Energy Units (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher leads a whole-class Claim–Evidence–Reasoning (CER) discussion. Learners look at the class data table projected on the board (teacher quickly enters representative pair values). Together, the class notices: (1) KE scales with mass linearly; (2) KE scales with speed squared (the quadrupling pattern). Teacher guides learners to write the proportionality: KE $\propto$ mv^2. A brief historical note is shared: 'Scientists in Europe in the 1600s and 1700s — including Gottfried Leibniz — debated exactly this relationship for decades. In Kenya, engineers who design the Nairobi Expressway use this same equation to calculate the energy of vehicles at speed.' The constant ½ is introduced via teacher explanation (derivation from calculus is acknowledged but deferred). For PE, learners identify that PE $\propto$ mgh from their data, and the equation is formalised. Learners then solve two contextualised problems in their exercise books: (Problem 1) A 70 kg matatu tout runs at 4 m/s to catch his vehicle — calculate his KE. (Problem 2) A 5 kg bag of maize is lifted 2 m onto a grain store shelf in Kitale — calculate its PE ($g = 10\ m/s^2$). Pairs cross-check answers.	1. Project the class data table. Say: 'We are now going to be detectives. Our data is our evidence. Let's write our claim.' Write on the board: 'CLAIM: Kinetic energy depends on _____ and _____.' Cold-call 2 learners to fill in the blanks. Apply 10-second WAIT TIME. 2. Say: 'Now, EVIDENCE. Point to the specific numbers in your data table that support this claim. Tell me the row numbers.' Cold-call 1 learner, ask them to read the specific data values aloud. 3. Say: 'Now REASONING — WHY do those numbers support the claim? How does the pattern in the numbers show us the relationship?' Give 3 minutes pair discussion, then cold-call 2 learners. 4. Write: KE $\propto$ mv^2 on the board. Say: 'Your data gave us this. Scientists confirmed it. The constant is ½ — we will revisit why in a later lesson. So: KE = ½mv^2.' Write the equation large and boxed. 5. Repeat the CER process for PE (shorter, 4 minutes): 'Our Kisumu video and your data both show PE depends on what three things?' Cold-call 3 learners. Write PE = mgh. 6. Say: 'Now apply it.' Read Problem 1 aloud slowly. Say: 'Work silently for 3 minutes.' Circulate, note errors (common: forgetting the ½, squaring the mass instead of velocity). After 3 minutes, ask a learner to write their solution on the board. Ask the class: 'Do you agree? If not, put your hand up and tell us which step is wrong.' 7. Repeat for Problem 2. 8. Explicitly bridge to phenomenon: 'The spanner arm — as it swings — has KE. Before the mechanic pushes, it has PE due to its raised position. After the nut loosens, where has that energy	Claim–Evidence–Reasoning (CER) Framework: CER forces learners to separate observation (evidence) from interpretation (reasoning) and conclusion (claim), a core science and engineering practice. By having learners construct the equations from their own data before the teacher formalises them, the mathematical relationships feel discovered rather than dictated. The contextualised problems (matatu tout, maize bag in Kitale) ensure immediate transfer to Kenyan everyday contexts, reinforcing that physics describes the world learners already live in.	OBSERVABLE INDICATOR: All learners have written KE = ½mv^2 and PE = mgh in their exercise books, boxed, with units labelled (Joules). Both worked problems are attempted with workings shown. Teacher checks specifically: (a) Is the ½ included in Problem 1? (b) Is $g = 10$ used correctly in Problem 2? EXIT TICKET (last 2 minutes of this phase): Teacher says: 'Write one sentence: What is the difference between kinetic energy and potential energy? Use the words mass, speed, and height.' This is collected or self-assessed against a model sentence displayed after 90 seconds.

		gone? We will find out. For now, you have the tools to measure it.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Conservation of energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** SI Kinetic Energy Units (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are introduced formally to the Driving Question Board. The teacher posts a large printed photograph of the Gikomba jua kali mechanic at the centre of the board with the unit driving question written above it: 'Why can a jua kali mechanic loosen a wheel nut effortlessly with a long spanner when a short one fails — and where does the energy go when friction makes the nut hot?' Each learner receives two sticky notes (or small pieces of paper). On the FIRST sticky note (blue/light colour), they write one thing they NOW UNDERSTAND about energy after today's lesson. On the SECOND sticky note (yellow/different colour), they write one 'wonder question' — something today's lesson made them curious about that they cannot yet answer. Learners post both notes on the DQB around the central phenomenon image. The teacher reads 5–6 wonder questions aloud and groups them roughly into two clusters: 'Questions about force and machines' and 'Questions about where energy goes (heat, sound, etc.)'. These clusters will be explicitly answered in subsequent lessons.	1. Unveil the pre-prepared DQB (a section of the classroom noticeboard or a large sheet of paper on the wall). Say: 'This board is the story of our investigation. Every week, we will add to it, revise it, and watch our understanding grow. By the end of this unit, every question on this board will have an answer — and YOU will have found those answers.' 2. Distribute sticky notes. Say: 'On the BLUE note: one thing you understand about energy that you did NOT understand at the start of today's lesson. Be specific — mention numbers or equations if you can. On the YELLOW note: one question today's lesson made you wonder about. It must connect to the mechanic at Gikomba. You have 4 minutes.' START timer. Circulate, read over shoulders, and privately encourage any learner who writes a very shallow wonder question: 'You wrote that you wonder why the nut gets hot. Good — can you make that question more specific? What about the energy?' 3. Ask learners to post their notes. While they do, teacher reads notes quickly. Select 5–6 yellow notes to read aloud. Say: 'Listen — these are YOUR questions. They are real scientific questions. Let me group them.' Move notes into two rough clusters on the board, narrating: 'These questions are about force and machines — we will investigate those in Lessons 3, 4, and 5. These questions are about energy disappearing as heat — we will crack that mystery in Lessons 2	Driving Question Board (DQB) as a Living Epistemic Artefact: The DQB externalises learners' evolving understanding and makes the unit's intellectual journey visible. By distinguishing 'what I now understand' from 'what I still wonder,' learners practise metacognition and experience science as an ongoing, unfinished process rather than a set of facts to memorise. The teacher's act of clustering wonder questions into themes shows learners that scientific inquiry is organised — curiosity is structured, not random. This directly supports the NGSS practice of Asking Questions and Defining Problems.	OBSERVABLE INDICATOR: Every learner has posted both a blue (understanding) and yellow (wonder) sticky note. Teacher scans blue notes for evidence that learners can articulate KE or PE in their own words (not just copy the equation). RED FLAG: If blue notes say only 'KE = $\frac{1}{2}mv^2$' without any explanatory language, this signals surface-level encoding — teacher makes a mental note to start Lesson 2 with a quick oral elaboration task ('Tell me what that equation means in words before you write it'). Wonder questions are assessed for connection to the phenomenon — notes that show no connection to the mechanic or energy transformation indicate a learner who has not yet engaged with the phenomenon and will be prioritised for individual check-in at the start of Lesson 2.

		and 6.' 4. Point to the driving question. Say: 'We cannot answer this yet. But today, you have built the first piece of the puzzle: you now know what energy is and how to measure it. Next lesson, we will investigate what happens to energy when it transforms — including into the heat that makes that nut warm.'		
Model Building Phase 📹 **VIDEO:** Conservation of energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** SI Kinetic Energy Units (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners create their first individual scientific model in their exercise books. The model task is: 'Draw the jua kali mechanic's spanner system at TWO moments — (A) the moment before the mechanic pushes (spanner arm raised), and (B) the moment the nut begins to move (spanner arm at lowest point). On your drawing, label: where KE is greatest, where PE is greatest, the direction of energy transfer, and add a question mark (?) on anything you cannot explain yet — including where the heat energy in the warm nut comes from.' Learners spend 6 minutes on this individual task. Pairs then compare models and note one similarity and one difference. Three learners share their models under the document camera (or hold them up). The class identifies what all three models have in common as a 'consensus model feature' — written on the board as the unit's first shared model statement: 'Before the push: PE is high, KE is low. After the push: KE is high, PE is low. Something still unknown happens to produce heat.'	1. Say: 'Scientists don't just calculate — they build models. A model is a drawing or diagram that shows how you think something works. Your model does not need to be perfect — it needs to be HONEST about what you know and what you don't know yet. Question marks are allowed. Question marks are encouraged.' 2. Write the model task on the board and read it aloud. Say: 'You have 6 minutes. Work individually in silence.' START timer. Circulate and look for: (a) labelling of KE and PE in correct positions; (b) an arrow or symbol showing energy transfer direction; (c) a question mark on the heat/friction element. Gently prompt any learner who draws only the mechanic without labelling energy: 'Where is the energy stored before he pushes? Can you show that on your drawing?' 3. After 6 minutes: 'Turn to your partner. Find ONE thing your models agree on and ONE thing that is different. You have 2 minutes.' 4. Cold-call 3 learners to share models (use document camera if available, otherwise ask them to describe their model while others follow along). After each, ask the class: 'What is correct in this model? What might need to change as we learn more?' Apply 10-second WAIT TIME. 5. Write	Progressive Model Building (Anchored Modelling): By building a model in Lesson 1 that explicitly includes question marks and unknowns, the teacher normalises scientific uncertainty and frames all subsequent lessons as model-revision opportunities. The act of labelling KE and PE on the spanner diagram directly connects the abstract equations just derived to the concrete phenomenon, making the mathematics physically meaningful. The consensus model statement gives the class a shared intellectual starting point — a claim they all own — which creates cognitive dissonance when Lesson 2 introduces energy dissipation and forces a revision.	OBSERVABLE INDICATOR: Each learner's Model Version 1 in their exercise book contains: (1) two labelled diagrams (before and after push); (2) KE and PE labels in correct positions (PE high before push, KE high at point of motion); (3) at least one question mark indicating an unexplained element (ideally on the heat/friction aspect). Teacher collects or photographs 5 randomly selected models at the end of the lesson for rapid review before Lesson 2. If fewer than 60% of sampled models correctly place PE before the push and KE during motion, teacher begins Lesson 2 with a 5-minute 'model check' discussion before proceeding to new content.

		the consensus model statement on the board. Say: 'This is our starting model. We will revise this model in EVERY lesson. By Lesson 12, it will look completely different — and much more powerful. Keep this first version so you can see how far you've come.' 6. Direct learners to write the consensus statement under their model drawing and label it: 'Model Version 1 — Lesson 1.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions in your professional journal: (1) Did learners' initial predictions about the sprinting athlete vs. the falling stone reveal genuine prior knowledge about mass and velocity, or did most learners rely on intuition about 'strength'? How will this inform your questioning in Lesson 2? (2) During the PhET simulation activity, did learners notice the quadrupling relationship between speed and KE independently, or did it require significant scaffolding? If heavy scaffolding was needed, consider pre-teaching ratio reasoning with a short number pattern exercise at the start of Lesson 2. (3) Were the wonder questions on the DQB genuinely connected to the mechanic phenomenon, or were they generic ('I wonder what energy is')? If they were generic, the phenomenon introduction may need a more vivid, visceral re-presentation at the start of Lesson 2 — consider playing a short audio clip of the mechanic's grunting effort before the pipe extension and then the ease after, to create stronger emotional anchoring. (4) Did all learners — including girls and learners from non-science backgrounds — feel safe to post honest wonder questions, or did social dynamics suppress uncertainty? Review whether a private submission method (folded paper rather than public sticky notes) would better serve psychological safety in this class. (5) How did your cold-call choices reflect equity? Did you call on learners from across gender, seating position, and perceived ability? Note three specific learners to prioritise for cold-call in Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	A jua kali mechanic in Gikomba, Nairobi, loosening a stubborn matatu wheel nut effortlessly with a long spanner-pipe extension after failing with a short spanner; a wheel nut that feels warm after repeated failed attempts; a Kenyan sprinter running at full speed; a stone falling from a grain-store roof in Kitale; PhET Energy Skate Park simulation showing energy bars changing with mass, speed, and height; a Kisumu construction-site video showing a stone making deeper dents when dropped from greater heights.
What did I learn?	Kinetic energy (KE) is the energy an object has due to its motion: $KE = ½mv^2$. Gravitational potential energy (PE) is the energy stored in an object due to its height above a reference point: $PE = mgh$. KE increases with both mass and the square of speed — doubling speed quadruples KE. PE increases linearly with both mass and height. Both are measured in Joules (J). In the mechanic's spanner system, the raised spanner arm stores PE and converts it to KE as it swings down. The heat in the wheel nut represents energy that has gone somewhere — but we cannot yet fully explain where.
How does this explain the phenomenon?	Using $KE = ½mv^2$, a 70 kg matatu tout running at 4 m/s has $KE = ½ \times 70 \times 4^2 = 560$ J. Using $PE = mgh$, a 5 kg bag of maize lifted 2 m onto a grain-store shelf in Kitale has $PE = 5 \times 10 \times 2 = 100$ J. The spanner arm before the mechanic's push stores gravitational PE (high position); as it swings, this converts to KE. The warm wheel nut is evidence that some energy is transformed into thermal energy — a mystery that points directly to the unit's driving question about energy conservation and dissipation, which

	will be fully resolved in subsequent lessons.

LESSON 2 (80 minutes): Conservation of Mechanical Energy: The Invisible Transfer	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate how kinetic energy (KE) and potential energy (PE) continuously interchange in a mechanical system, and to establish that total mechanical energy remains constant in the absence of friction.
Knowledge	Learners will be able to: (1) define kinetic energy as energy possessed by an object due to its motion ($KE = ½mv^2$) and potential energy as energy stored due to an object's position ($PE = mgh$); (2) state the principle of conservation of mechanical energy; (3) describe the continuous interchange between KE and PE in systems such as a pendulum and a roller coaster.
Skills	Learners will be able to: (1) use the PhET Pendulum Lab simulation to gather quantitative evidence of KE–PE interchange; (2) construct a class energy transformation model diagram showing energy flow at different points in a pendulum's swing; (3) analyse a graphical representation of KE and PE vs. time to identify patterns of conservation.
Attitudes	Learners will appreciate that natural systems follow consistent, predictable rules (energy conservation), fostering curiosity about why real systems (like the jua kali mechanic's spanner struggle) appear to 'lose' energy — motivating deeper inquiry into friction and heat.
Key Inquiry Question	How is energy conserved and transformed in mechanical systems?
Purpose in Storyline	In Lesson 1, learners established that the jua kali mechanic does work when applying force through the spanner. This lesson deepens that foundation: learners now discover that mechanical energy is not destroyed — it only transforms between KE and PE. This directly sharpens the unit's second mystery: if energy is conserved, why does the wheel nut get hot and where does the 'lost' energy go? The lesson positions friction and heat as the next evidence target (Lesson 3).
Safety Notes	This lesson uses digital simulations and video only — no physical hazards. Ensure all devices are charged or plugged in. If using a physical pendulum demonstration, keep the swing arc small and ensure no learners stand within the pendulum's path. Remind learners never to touch electrical outlets or extension cords when setting up devices.

B. LESSON OVERVIEW
Learners use the PhET Pendulum Lab simulation and a YouTube roller-coaster/pendulum demonstration to gather evidence that kinetic energy (KE) and potential energy (PE) continuously interchange in a frictionless mechanical system, and that their sum — total mechanical energy — remains constant. Starting from the Lesson 1 anchor (the jua kali mechanic exerting force through a spanner), learners are challenged: if the mechanic puts energy INTO the system, where does it go? A guided simulation activity, graphical analysis, and a whole-class energy transformation model diagram build understanding progressively. The lesson closes by updating the Driving Question Board with a sharpened puzzle: 'But if energy is always conserved, why does the wheel nut get HOT?'

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Intro to the	Learners are shown a projected still image: a steel ball on a string (pendulum) held at an angle at	Display the still pendulum image and say: 'Before we touch anything, I want YOUR thinking on	Predict-before-observe (elicitation of prior models): By committing predictions to paper before seeing	Observable indicator: All learners produce a labelled sketch with at least two of the three positions

endocrine system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Energy Conservation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	shoulder height by a student volunteer in the classroom — positioned to represent the mechanic's arm raised before pushing the spanner. The teacher asks: 'I am going to release this ball. Sketch on your whiteboard/notebook what you think the ball's speed and height will look like at THREE points: (A) the moment of release, (B) the lowest point of the swing, and (C) the far side at the top of the swing. Label which point has the MOST energy of motion and which has the MOST stored energy.' Learners work individually for 3 minutes, sketching and labelling. Pairs then share predictions for 2 minutes. The teacher cold-calls 4 pairs to share, recording key prediction words (fast, slow, high, low, stored, moving) on the board in two columns: MOTION ENERGY | STORED ENERGY.	the board. You have 3 minutes — sketch, label, predict.' [WAIT — do not answer questions for 3 minutes; circulate silently, noting diverse predictions.] After 3 minutes: 'Turn to your neighbour. Compare your sketches. Do you agree on which point is fastest?' [WAIT 2 minutes.] Cold-call 4 pairs using name sticks: 'Amina and Otieno — what did you predict at the lowest point?' Record responses verbatim. Then say: 'Hold those predictions. We are going to test them RIGHT NOW with evidence.'	evidence, learners activate existing mental models of motion and stored energy. Recording prediction language on the board in two columns (MOTION ENERGY | STORED ENERGY) primes learners to use formal vocabulary (KE, PE) when it is introduced in the Explain phase, creating a cognitive anchor for new terminology.	annotated. Teacher scans for common misconceptions: (1) learners who predict the ball moves fastest at the top — flag for targeted questioning in Phase 3; (2) learners who draw the ball as equally fast everywhere — use as a discussion contrast case. Record 2–3 representative misconception sketches on the board for later revisiting.
Observe Phase 📹 **VIDEO:** Intro to the endocrine system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Energy Conservation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	ACTIVITY A — PhET Pendulum Lab Simulation (25 min): Learners open the PhET 'Pendulum Lab' simulation (pre-loaded on school tablets/laptops or accessed via the ARES offline player). Working in pairs, they follow a structured evidence-gathering guide: (1) Set pendulum length to 1.0 m, mass to 1.0 kg, and friction to ZERO. (2) Pull the bob to 30° and release. Observe the motion for 10 full swings. (3) Open the 'Energy' graph panel — observe the KE (blue) and PE (red) lines over time. Record: 'When KE is at its PEAK, where is the pendulum?' and 'When PE is at its PEAK, where is the pendulum?' (4) Freeze the simulation at three moments: release point, lowest point, opposite peak — read and record	Distribute the structured evidence-gathering guide before opening devices. Say: 'Your job right now is NOT to understand everything — your job is to be a detective. Write down EXACTLY what the numbers tell you. Do not skip the SUM column — that column is the whole point of today.' [WAIT — circulate for 10 minutes, crouching to pair level.] At the 10-minute mark, pause the class: 'Raise your hand if your SUM column numbers are roughly the SAME at all three points.' [Count hands aloud.] 'Interesting. Raise your hand if they are DIFFERENT.' Address outliers: 'Check — did you set friction to zero? Recalculate.' Before the video, say: 'As you watch, your only job is to annotate those five points on the roller	Evidence-recording with a structured data table (Science and Engineering Practice: Analysing and Interpreting Data): The deliberate inclusion of a SUM column transforms a passive simulation exploration into a quantitative test of a hypothesis (Is total energy constant?). The SUM column makes the conservation principle visible as a pattern in numbers, not just a statement to memorise. The roller-coaster extension applies the pattern to a second context, building generalisability.	Observable indicator: Each pair produces a completed data table with KE, PE, and SUM values at three pendulum positions, plus two written sentences describing the effect of friction on total energy. Teacher checks for: (1) SUM values that are approximately equal (within ~0.1 J) at all three frictionless positions — correct; (2) SUM values that decrease when friction is turned on — correct and critical for the next lesson; (3) pairs who report that KE and PE are BOTH zero simultaneously — prompt: 'Can the pendulum be motionless AND at zero height at the same time? Look again.'

	KE and PE values in a table. (5) Calculate KE + PE at each point and record the SUM. (6) Now INCREASE friction to maximum. Observe what happens to the total energy sum over time. Record observations in 2 sentences. ACTIVITY B — YouTube Roller Coaster Demonstration (5 min): Teacher plays a 3-minute segment of a roller-coaster energy video (pre-downloaded, played offline). Learners annotate a printed/projected roller-coaster outline at 5 marked points, writing KE HIGH / PE HIGH / BOTH MEDIUM at each point, connecting back to the PhET evidence.	coaster. Pause your thinking — just label.' After the video: 'In one sentence, tell your partner what the roller coaster and the pendulum have in common.' Cold-call 3 pairs.		
Explain Phase 📹 **VIDEO:** Intro to the endocrine system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Energy Conservation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Vocabulary Anchoring (5 min): Teacher introduces formal definitions using the students' own prediction language from Phase 1. Projecting the two-column board list (MOTION ENERGY | STORED ENERGY), the teacher writes KE = ½mv² next to 'motion energy' and PE = mgh next to 'stored energy', narrating: 'You already had the right idea — we are just giving it a precise name and a precise formula.' Learners copy definitions with a worked example: 'A matatu wheel of mass 12 kg rolling at 4 m/s — what is its KE?' (Answer: ½ × 12 × 16 = 96 J). PART B — Class Energy Transformation Model Diagram (15 min): The teacher draws a large pendulum arc on the board. The class co-constructs the model: teacher calls on individual students (cold-calls, minimum 6 students) to come forward and annotate one feature each — arrow direction of energy flow, label KE/PE at each point, write the conservation equation $KE_1 + PE_1 = KE_2 + PE_2$. Each	During vocabulary anchoring, point explicitly to the board: 'Do you see? You said MOTION ENERGY in your predictions. Scientists call it kinetic energy. Same idea — now we can calculate it.' For the model diagram, use name sticks to cold-call: 'Wanjiru, come and draw an arrow showing which direction energy is flowing as the pendulum rises. Explain to us why.' [WAIT 10 seconds after each student speaks before responding or prompting the next.] After 6 annotations, step back and ask the class: 'Is there anything on this diagram that doesn't match your evidence from the simulation? Anyone disagree with what's written?' [WAIT 15 seconds — silence is acceptable and productive.] For the phenomenon connection, say: 'Write three sentences. Sentence 1: where is the energy stored when the mechanic raises the spanner. Sentence 2: where does it go when he pushes. Sentence 3: what does this tell us about where the EXTRA force comes from.'	Co-constructed annotated model diagram (Science and Engineering Practice: Developing and Using Models): Students physically walk to the board and annotate, making the model a shared class artefact rather than teacher-transmitted information. This externalises reasoning, allows peer critique, and produces a reference diagram that remains on the board for all subsequent lessons. Connecting the model back to the phenomenon (the mechanic's arm) ensures that abstract KE/PE concepts are immediately grounded in the anchoring context.	Observable indicator: Each learner produces a 3-sentence written explanation connecting KE/PE to the jua kali phenomenon. Teacher reads 5 randomly selected responses during the DQB phase. Success criterion: sentences must use the words 'kinetic energy,' 'potential energy,' and one conservation statement ('energy is not created or destroyed'). Flag learners who write that 'the long spanner creates more energy' — this is a key misconception to address explicitly in Lesson 3 (mechanical advantage does not create energy).

	annotator must justify their placement in one sentence. PART C — Connecting to the Phenomenon (5 min): Teacher projects the original jua kali mechanic image and asks: 'The mechanic lifts his arm (with the long spanner) up before pushing down. At that highest point — is this KE or PE? As his arm pushes down and the spanner moves, what happens to that energy?' Learners write a 3-sentence explanation in their notebooks connecting the pendulum model to the spanner system.	[Give 4 minutes of individual writing time.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Intro to the endocrine system] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Energy Conservation (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher uncovers the class Driving Question Board (a large sheet of paper or section of the whiteboard that has been maintained since Lesson 1). The existing DQB entries from Lesson 1 are visible (e.g., 'What is work?', 'How does a longer spanner help?'). Learners are each given two sticky notes. On NOTE 1 (green): they write one thing they NOW understand about the driving question that they didn't understand before this lesson. On NOTE 2 (orange): they write one NEW question raised by today's evidence — specifically, something that is still puzzling. Learners post their sticky notes on the DQB under two columns: WHAT WE NOW KNOW | WHAT WE STILL WONDER. The teacher reads 4–5 orange notes aloud. The class votes (show of hands) on which new question is MOST puzzling. The teacher circles the winning question — expected to be a variation of: 'If energy is always conserved, why does the wheel nut get hot? Where does the heat come from?' — and	Uncover the DQB with deliberate ceremony: 'Let's revisit the board we started last lesson. Read the questions we left here.' [Give 30 seconds of silent reading.] Distribute sticky notes: 'Green note — one thing you NOW know. Orange note — one NEW question today's evidence raised. You have 3 minutes. Go.' [Circulate, prompt quiet students: 'What surprised you most in the simulation?'] After posting, read 5 orange notes aloud without commentary. Then: 'Hands up — which of these questions would you MOST want to investigate next?' Count hands visibly. Circle the winning question and say: 'This is not a mistake — this is exactly how scientists work. Today's answer creates tomorrow's question. That question about heat and energy disappearing? That is Lesson 3.'	Driving Question Board with colour-coded sticky notes (Science and Engineering Practice: Asking Questions and Defining Problems): The two-colour system explicitly separates consolidated knowledge (green) from productive uncertainty (orange), making the iterative nature of scientific inquiry visible as a physical artefact. The class vote democratises the inquiry direction and increases buy-in for subsequent lessons.	Observable indicator: Every learner posts both a green and an orange sticky note. Teacher scans green notes for quality: acceptable responses reference KE, PE, or conservation; unacceptable responses are vague ('I learned about energy'). Orange notes are the primary diagnostic — teacher photographs the DQB after class to analyse the distribution of new questions and use them to inform the Lesson 3 introduction. Specifically note any learners who write that 'energy is destroyed by friction' — this reveals the critical misconception the unit must address.

	announces: 'That question is EXACTLY where we are going next.'			
Model Building Phase 📹 **VIDEO:** Intro to the endocrine system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Energy Conservation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their PERSONAL UNIT MODEL — a running diagram begun in Lesson 1 that will accumulate evidence across all 8 lessons. In Lesson 1, learners drew the jua kali mechanic applying force through the spanner. Now, they ADD to that model: (1) Draw energy flow arrows showing PE → KE as the mechanic's arm swings the spanner downward. (2) Annotate with the conservation equation: $KE_1 + PE_1 = KE_2 + PE_2$. (3) Add a QUESTION BUBBLE (drawn as a cloud shape) near the wheel nut with the text: 'Heat? Where does this energy go?' — explicitly flagging the unresolved mystery. (4) Learners who finish early are challenged to add a second scenario: annotate a matatu tyre rolling downhill on the Nairobi–Nakuru highway — where is KE highest? Where is PE highest? Pairs share models briefly (2 min), then models are stored in their science portfolios.	Project the Lesson 1 model (a class photograph of the model from last lesson) alongside today's co-constructed class diagram. Say: 'Your personal model needs to GROW today. Open to your unit model page. You are adding three things — energy flow arrows, the conservation equation, and a question bubble for the heat mystery. You have 6 minutes.' [Circulate, prompting: 'Show me where PE is highest in your mechanic diagram. Now show me where KE is highest.'] At the 5-minute mark: 'If you are done, add the matatu on the Nairobi–Nakuru road challenge at the bottom.' After sharing: 'Your model is not finished — it will never be finished until Lesson 8. Every lesson, we add more. Today you added energy flow. Next lesson, you will add something about what happens to energy when friction is involved.' Collect models to check for the three required additions.	Cumulative personal model revision (Science and Engineering Practice: Developing and Using Models; Science and Engineering Practice: Constructing Explanations): The running personal model forces learners to synthesise each lesson's evidence into a single growing representation. The question bubble protocol normalises productive uncertainty — learners are explicitly asked to represent what they do NOT yet know, making the model an honest reflection of their current understanding rather than a polished final answer. The Nairobi–Nakuru extension contextualises PE/KE interchange in a familiar Kenyan road geography.	Observable indicator: Teacher collects models at the end of class and checks for three specific additions: (1) at least one PE→KE arrow on the spanner diagram; (2) the conservation equation written anywhere on the model; (3) a question bubble referencing heat or 'missing' energy near the wheel nut. Models missing any of these three elements are returned with a written prompt at the start of Lesson 3: 'Before we begin — add this to your model.' Track which learners are missing the question bubble specifically, as this may indicate low engagement with the unresolved mystery that drives the unit forward.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) EVIDENCE QUALITY: Did learners' PhET data tables show approximately equal SUM values across all three frictionless pendulum positions? If not, was this a simulation error, a setup error (friction not set to zero), or a conceptual misunderstanding of which values to read? (2) MISCONCEPTION TRACKING: How many learners wrote on their green sticky note that 'the longer spanner creates more energy'? This misconception — confusing mechanical advantage with energy creation — must be directly addressed in Lesson 3's opening. Prepare a targeted 3-minute rebuttal using the PhET data the class generated today. (3) DQB QUALITY: Did the winning orange question explicitly reference heat and energy? If learners' most-voted question was about force or the length of the spanner, the phenomenon connection to heat dissipation needs re-anchoring at the start of Lesson 3 using a stronger stimulus (e.g., a 30-second video clip of a mechanic feeling a hot bolt). (4) MODEL BUILDING: Were learners able to independently add energy flow arrows to the mechanic diagram, or did most copy the class model without personalising it? In Lesson 3, introduce a 'different from the class model' challenge to push individual sense-making. (5) PACING: The PhET activity is the highest-risk phase for time overrun. If the class needed more than 25 minutes on Phase 2, consider splitting the roller-coaster video into a 5-minute starter for Lesson 3 rather than including it here.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Using the PhET Pendulum Lab simulation, learners observed that as a pendulum swings, kinetic energy (KE) is highest at the lowest point and potential energy (PE) is highest at the two extreme ends — and that the SUM of KE + PE remained approximately constant across all positions when friction was set to zero. When friction was introduced, the total energy SUM decreased over time. Learners also annotated a roller-coaster diagram at five points, identifying where KE and PE were highest.
What did I learn?	Learners learned that kinetic energy (KE = $\frac{1}{2}mv^2$) is the energy of motion and potential energy (PE = mgh) is stored energy due to position. In a frictionless system, KE and PE continuously interchange but their total — mechanical energy — is conserved. This is expressed as $KE_1 + PE_1 = KE_2 + PE_2$. When friction is present, the total mechanical energy decreases — raising the new question of where that energy goes.
How does this explain the phenomenon?	Learners applied the conservation of mechanical energy to the jua kali mechanic phenomenon: when the mechanic raises his arm (and the long spanner) before pushing, he stores potential energy in the system. As the spanner swings downward, that PE converts to KE — delivering force to the wheel nut. The class energy transformation model diagram shows this flow. However, the wheel nut getting hot remains unexplained by conservation of mechanical energy alone, anchoring the next investigation into friction and thermal energy.

LESSON 3 (80 minutes): Work: Definition and Calculation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To formally define work in the scientific sense and apply the equation W = Fd cosθ to calculate work done in real mechanical situations, including the jua kali spanner scenario.
Knowledge	Learners should be able to: (a) define work as the product of force and displacement in the direction of the force; (b) state and apply the formula W = Fd cosθ; (c) distinguish between positive work, negative work, and zero work; (d) explain how friction dissipates energy as heat during mechanical work.
Skills	Learners should be able to: (a) measure force (using a spring balance or estimated body weight) and displacement during a desk-pushing activity; (b) calculate work done using W = Fd cosθ; (c) compare work done by the mechanic with a short versus long spanner handle; (d) represent energy transfers using annotated diagrams.
Attitudes	Learners should: (a) appreciate that scientific definitions of everyday words (like 'work') can differ from common usage; (b) value precision in measurement and calculation; (c) show curiosity about why friction converts useful mechanical energy into heat.
Key Inquiry Question	How is energy conserved and transformed in mechanical systems?
Purpose in Storyline	In Lessons 1 and 2, learners anchored their investigation in the jua kali mechanic phenomenon and explored forces and moments qualitatively. Lesson 3 formally introduces the quantitative concept of work (W = Fd cosθ), giving learners the mathematical tool to calculate and compare the energy inputs for the short and long spanner. This is the pivotal bridge between the qualitative force observations of earlier lessons and the energy-conservation and efficiency calculations that follow in Lessons 4–8. The warming of the wheel nut — puzzling since Lesson 1 — is now partially explained: friction does negative work on the nut, converting kinetic energy to thermal energy.
Safety Notes	Ensure desks are pushed on a clear, unobstructed floor — remove bags and cables from the path before the activity. Students should wear closed shoes. When using spring balances, do not exceed the rated scale. No sharp tools are used in this lesson.

B. LESSON OVERVIEW
Learners begin by revisiting their Lesson 1–2 predictions about the spanner scenario. The teacher re-anchors on the driving question and poses a new puzzle: the mechanic pushes hard but the nut barely moves — has he done any 'work'? Through a structured desk-pushing activity on the classroom floor, learners measure force and displacement to calculate work done, distinguish positive/negative/zero work, and reconnect to the phenomenon by calculating and comparing work done with a short versus long spanner. The lesson closes with a model revision that adds the W = Fd cosθ arrow-diagram notation to learners' cumulative anchor chart.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:**	Learners are shown a projected image of the jua kali mechanic at Gikomba market: in one frame he	Display the two-frame image and say: 'Look carefully at both frames. I want you to think about the word	Think-Pair-Share with Prediction Journaling: Learners externalise their prior conceptions of 'work' in	Circulate during silent writing and note: (a) how many learners correctly identify that zero

[Watch Me](#) Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** [Guess and Check, Work Backward (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	pushes the short spanner 5 cm without the nut moving; in the second frame he pushes the long spanner 15 cm and the nut turns. Each learner silently writes in their science notebook: (1) In which frame did the mechanic do more 'work' in the everyday sense? (2) In which frame did he do more 'work' in the scientific sense? (3) If he pushed the short spanner with 200 N of force but the nut never moved, how much work did he do scientifically? Learners then share predictions with a partner (Think-Pair-Share) before three cold-call shares to the class.	WORK — not the way your parents use it on a Monday morning, but what you think it means scientifically.' Allow 2 minutes silent writing. Then say: 'Turn to your desk partner. You have 90 seconds — share your three answers. Disagree where you disagree.' After pair talk, cold-call exactly 3 learners using name sticks: 'Akinyi, what did your pair decide about Frame 1?' — 'Kipchoge, do you agree or would you change anything?' — 'Fatuma, what about the nut that never moved — did the mechanic do scientific work?' After each response, wait 12 seconds before redirecting. Do NOT confirm or correct yet — say: 'Hold those ideas. We're going to test them today.'	writing before instruction, creating a personal baseline they will explicitly compare against after the activity. This surfaces the common misconception that effort = work and prepares cognitive conflict when the formal definition is introduced.	displacement = zero work (target: at least 40% before instruction); (b) whether learners conflate force with work. Use this to calibrate depth of the Explain phase. Observable indicator: learner notebooks contain a written prediction for all three prompts.
Observe Phase 📹 **VIDEO:** [Watch Me](#) Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** [Guess and Check, Work Backward (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	ACTIVITY: 'The Desk-Push Challenge.' Groups of 4 are each assigned one of three tasks: (A) Push a school desk 2 metres across the classroom floor with a steady horizontal force — one learner pushes, one reads the spring balance attached to the desk (or estimates using a bathroom scale held horizontally), one measures displacement with a metre ruler, one records. (B) One learner holds a heavy schoolbag stationary at shoulder height and walks 3 metres horizontally across the room — measure displacement and note that the lifting force is vertical but motion is horizontal. (C) One learner pushes the desk and a second learner (acting as 'friction') leans against the opposite side, resisting motion — record the direction of the friction force relative to displacement. Each group records:	Before the activity, briefly assign roles: 'Group A — you are the positive work team. Group B — you are the zero work team. Group C — you are the negative work team. Don't tell each other what those labels mean yet.' Circulate during the 12-minute activity. Prompt Group B: 'Which direction is the force holding the bag up? Which direction is the person walking? Draw an arrow for each.' Prompt Group C: 'What direction is friction acting compared to the direction the desk moves?' After groups finish recording, reconvene and say: 'Each group, report your three numbers: force, displacement, and angle. Write them on the board.' Pause 10 seconds. Then say: 'I notice Group B's force and displacement arrows are pointing in different directions — what angle is between them?' Wait 12 seconds for a volunteer.	Structured Role-Based Data Collection with Comparative Data Board: By assigning each group a qualitatively different scenario (θ = 0°, 90°, 180°) before teaching the formula, learners generate the empirical data that the formula must explain. The shared class data board creates a common evidence base that the Explain phase formalises — building understanding from evidence rather than from definition alone.	Check each group's recorded data table for: (a) correct identification of force direction and displacement direction; (b) correct angle estimate (Group A: ~0°, Group B: 90°, Group C: ~180°). Observable indicator: all four group members can point to the force arrow and displacement arrow on their diagram and state the angle between them. Flag any group that draws force and displacement as the same arrow — intervene before the Explain phase.

	the magnitude of force applied (N), the displacement (m), the angle between force and displacement (0°, 90°, or 180°), and their observed result (desk moved / bag moved / desk resisted). Groups then watch a 2-minute teacher-narrated video clip (or teacher live demonstration) showing the jua kali mechanic: short spanner handle = 0.15 m, force = 250 N, nut does NOT turn (displacement = 0); long spanner handle = 0.45 m, force = 80 N, nut turns through an arc of 0.45 m. Learners record these values in a 'Spanner Data Table' in their notebooks.	Then narrate the spanner video/demo, pausing to ask: 'Before I tell you, predict: with the short spanner, the nut never moved — what is the displacement of the nut?' Wait 10 seconds. Cold-call 2 learners.		
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Guess and Check, Work Backward (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher formally introduces W = Fd cosθ using the class data board as the scaffold. Learners apply the formula immediately to: (1) their own desk-push data (Group A), (2) the bag-carrying scenario (Group B — discovering W = 0 because cos 90° = 0), (3) the friction scenario (Group C — discovering W is negative because cos 180° = -1). Learners then work individually to calculate: (a) Work done by the mechanic with the short spanner (W = 250 × 0 × cos 0° = 0 J — nut never moved); (b) Work done with the long spanner (W = 80 × 0.45 × cos 0° = 36 J). They record both calculations and answer: 'Why did the nut feel warm after the failed short-spanner attempts, even though W = 0 J on the nut?' (Guided answer: the mechanic's muscles still consumed chemical energy; friction between spanner and nut converted this to thermal energy even with no net displacement of the nut — energy was dissipated, not usefully transferred.) Learners annotate a printed diagram of the spanner	Write W = Fd cosθ on the board and say: 'This formula must explain every number on our data board. Let's test it.' Point to Group A's data and ask: 'If force is 40 N, displacement is 2 m, and angle is 0 degrees — what is cos 0°?' Wait 10 seconds. Cold-call one learner. Write the calculation on the board step by step. Then point to Group B and say: 'Same formula. Angle is 90°. What is cos 90°? Use your calculator.' Wait 12 seconds. Ask: 'So what is the work done carrying the bag horizontally? Does that feel right?' Wait 10 seconds — allow cognitive dissonance to surface ('But they were tired!'). Acknowledge: 'You are right that your muscles worked hard. But scientifically, useful work on the bag was zero — your muscles did work fighting gravity, but that force was perpendicular to motion.' Then say: 'Now apply this to the spanner. Work alone for 3 minutes — calculate both cases.' After 3 minutes, pair-share answers, then cold-call 2 learners to present their spanner calculations. When a	Formula-to-Evidence Mapping with Cognitive Conflict Resolution: Rather than presenting W = Fd cosθ abstractly, the formula is immediately applied to the data the class generated. The zero-work bag scenario creates productive cognitive conflict (effort ≠ work), which is resolved by distinguishing biological energy expenditure from mechanical work done on an object. This directly advances the driving question about where energy 'goes' — it is not lost but transformed.	Collect individual spanner calculation notebooks. Look for: (a) correct substitution of d = 0 for the short spanner case; (b) correct answer of 36 J for the long spanner; (c) a written explanation linking friction to heat dissipation. Observable indicator: ≥75% of learners correctly calculate W = 0 J for the short spanner AND provide a sentence explaining the warming of the nut. Learners who conflate 'effort' with 'work' should be flagged for targeted questioning in Phase 4.

	showing: F arrow, d arrow, θ label, W value, and a 'heat lost to friction' wavy arrow.	learner presents W = 0 J for the short spanner, ask the class: 'Then why did the nut get hot? Talk to your partner for 60 seconds.' Cold-call Fatuma or another learner who has not yet spoken.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Guess and Check, Work Backward (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board displayed at the front of the room. Each learner writes one sticky note answering: 'What new evidence from today helps us answer: WHERE DOES THE ENERGY GO when friction makes the nut hot?' and a second sticky note for: 'What new question has today's lesson raised for you about the spanner, work, or energy?' Sticky notes are posted under the two DQB columns: EVIDENCE WE NOW HAVE and NEW QUESTIONS. The class reads the new evidence notes aloud (teacher selects 4–5) and groups them by theme (e.g., 'friction converts energy to heat', 'zero displacement = zero work', 'longer handle = more displacement = more work done'). Learners update their personal DQB tracker in their notebooks.	Point to the DQB and say: 'Three lessons ago, you wondered why the nut gets hot and where the energy goes. Today you calculated W = 0 J on the nut with the short spanner — but we know energy went somewhere. On your sticky note, explain where.' Allow 3 minutes silent writing. As learners post notes, read a selection aloud and say: 'I'm seeing a pattern here — several of you are writing about friction. Let's cluster these.' Physically group the sticky notes on the board. Then ask: 'Who has a NEW question today raised that we haven't answered yet?' Cold-call 3 learners. Add their questions to the board in a different colour. Say: 'These new questions are exactly what Lessons 4 and 5 will investigate — efficiency and power. Hold those questions.'	Collaborative DQB Synthesis with Thematic Clustering: The DQB is not a passive display — by physically clustering evidence notes, learners see their collective growing understanding mapped spatially. The two-column structure (evidence vs. new questions) reinforces that science is an ongoing inquiry, not a set of finished answers, and models scientific practice 6 (constructing explanations) and practice 8 (obtaining, evaluating, and communicating information).	Read all posted sticky notes before the next lesson. Observable indicators: (a) at least 80% of evidence notes correctly reference W = Fd cosθ, friction, or energy dissipation; (b) new-question notes show forward thinking (e.g., 'How much energy is wasted as heat?' or 'Is a longer spanner always better?'). Learners who post vague notes ('work is force times distance') without connecting to the phenomenon should receive a targeted verbal prompt at lesson start next session.
Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Guess and Check, Work Backward (Flexbook) Source: CK-12 🔍 Search ARES	Learners revise their cumulative anchor-chart model of the jua kali spanner system. They must now ADD to their diagram from previous lessons: (1) a labelled arrow for the applied force F (in Newtons), a labelled arrow for displacement d (in metres), and the angle θ between them; (2) a calculation box showing W = Fd cosθ for both the short and long spanner scenarios; (3) a 'heat energy' wavy arrow at the nut-spanner contact point, labelled 'energy dissipated by friction (negative work)'; (4) a statement:	Say: 'Open your anchor-chart model from Lesson 2. Today you have three new tools: the W = Fd cosθ formula, the concept of positive/negative/zero work, and the idea that friction does negative work and converts energy to heat. Your job in the next 10 minutes is to make your model SHOW these three things — not just write them as labels, but draw them as part of the spanner diagram.' Circulate and prompt: 'Where exactly does the energy go when friction is at work? Can you show that with a wavy arrow?' and 'Your model	Cumulative Annotated Model Revision with Gallery Critique: The anchor-chart model is a living document that grows with each lesson, giving learners a visual record of their conceptual development. The gallery critique (NGSS Science and Engineering Practice 7: Engaging in argument from evidence) requires learners to evaluate another group's model against the evidence gathered today, deepening understanding through articulation and peer challenge.	Collect or photograph each group's revised anchor-chart model. Observable indicators: (a) the model includes correctly labelled F, d, and θ for at least one spanner scenario; (b) a wavy arrow or equivalent symbol represents friction-dissipated energy; (c) the energy-conservation statement ('input = useful work + heat lost') appears explicitly. Models lacking the friction/heat component should be flagged — this misconception (energy disappears rather than transforms) is the central

for similar readings	'Total energy input by mechanic = useful work done on nut + energy lost to friction.' Groups compare their revised models with another group and identify one difference and one similarity. Each learner writes a two-sentence 'model update statement' in their notebook explaining what their model can now show that it could NOT show after Lesson 2.	shows two spanner lengths — which calculation box shows more work done? Why?' After 10 minutes, pair models with a neighbouring group for a 3-minute gallery critique. Ask one spokesperson per pair to share: 'What was the most important addition your partner group made that you had missed?' Cold-call 3 pairs. Close by saying: 'Your model is getting more powerful. In Lesson 4, we'll add one more question: if the same work gets done faster — or slower — does that matter? That's power.'		conceptual target of the entire unit.

D. TEACHER REFLECTION
After the lesson, reflect on the following: (1) Did learners successfully distinguish between everyday 'work' (effort) and scientific work ($W = Fd \cos\theta$)? What percentage of the class correctly calculated W = 0 J for the short-spanner case? If fewer than 60% succeeded, plan a 10-minute re-engagement at the start of Lesson 4 using a simpler context — e.g., a market porter at Gikomba carrying goods on their head (zero work done on goods if force is vertical and motion is horizontal). (2) Did the DQB sticky notes show evidence of conceptual progression — specifically, did learners connect friction to energy dissipation rather than energy 'disappearing'? If not, revisit the first law of thermodynamics informally at the start of Lesson 4. (3) Were Group B learners (the bag-carrying group) genuinely surprised by W = 0 J? Productive confusion here is a teaching asset — note which learners voiced this conflict for targeted extension in Lesson 5. (4) Did the model-building phase produce models that included both the quantitative (formula, numbers) and qualitative (wavy heat arrow) components? If models remained purely verbal, introduce a model-drawing scaffold in Lesson 4. (5) Equity check: were female learners equitably cold-called and given wait time? Review your cold-call tally from today's lesson before planning Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students pushed a school desk across the classroom floor, held a heavy schoolbag while walking horizontally, and resisted a desk push to simulate friction. They also observed data from the jua kali spanner scenario (short handle: F = 250 N, d = 0 m; long handle: F = 80 N, d = 0.45 m) and recorded force, displacement, and angle for each scenario.
What did I learn?	Work is defined as $W = Fd \cos\theta$ — the product of force, displacement, and the cosine of the angle between them. When the angle is 0°, maximum positive work is done (force and motion in the same direction). When $\theta = 90°$, work done is zero (e.g., carrying a bag horizontally). When $\theta = 180°$, work is negative (friction opposes motion). The mechanic did zero scientific work with the short spanner because the nut did not move (d = 0), but 36 J of work with the long spanner. Friction converts mechanical energy to thermal energy — this is why the nut got warm.
How does this explain the phenomenon?	The jua kali mechanic's short spanner failed not because of insufficient force alone, but because the nut's displacement was zero — so no work was done on the nut regardless of force applied. The mechanic's biological energy was still consumed, but converted to heat by friction at the nut-spanner interface rather than to useful rotational work. With the long spanner, a smaller force over a greater displacement arc achieved the required work (36 J) to overcome the nut's resistance — illustrating that work

	depends on both force AND displacement in the direction of the force.

LESSON 4 (40 minutes): Work Done Against Friction: Where Does the Energy Go?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of this lesson learners will be able to measure the friction force on a sliding object, calculate the work done against friction, and explain how mechanical energy is converted to thermal energy — directly resolving why the wheel nut in the jua kali scenario heats up during repeated failed loosening attempts.
Knowledge	Learners will state that work done against friction equals the friction force multiplied by the distance slid (W_f = f_r x d); explain that this work is not lost but converted to thermal energy in the surfaces in contact; and distinguish between useful work output and energy dissipated as heat.
Skills	Learners will use a spring balance to measure friction force on surfaces of different roughness; record and tabulate data accurately; calculate work done against friction; and update a class energy-transformation model diagram to include an energy-dissipation arrow.
Attitudes	Learners will show curiosity about everyday energy puzzles such as why surfaces warm up during sliding, and appreciate that careful measurement can reveal invisible energy transfers happening in familiar Kenyan workplaces.
Key Inquiry Question	How is energy conserved and transformed in mechanical systems? (This lesson addresses the dissipation side: mechanical energy is conserved overall but some is transformed irreversibly to thermal energy through friction, so the total energy is still accounted for.)
Purpose in Storyline	Lessons 1 and 2 established that mechanical energy is conserved in frictionless systems. Lesson 3 introduced W = Fd cosθ and revealed that friction does negative work. This lesson quantifies that negative work, explains the thermal energy produced, and updates the class model with an energy-dissipation leak — setting up Lesson 5 (power) and Lesson 7 (efficiency) where students will compare useful work output with total work input.
Safety Notes	Ensure spring balances are not overloaded beyond their rated capacity. Sandpaper edges can cause minor abrasions; learners should slide the block away from their fingers. Wooden blocks must be free of splinters. No chemical or electrical hazards in this activity.

B. LESSON OVERVIEW
Learners begin by predicting whether the wheel nut heats up more when the mechanic uses the short or the long spanner handle — and why. They then conduct a hands-on investigation pulling a wooden block across smooth tile and sandpaper surfaces with a spring balance, measuring friction force and calculating work done against friction. They feel the surfaces after repeated sliding to detect warmth, directly connecting the measurement to the thermal effect. In the Explain phase they formalise W_f = f_r x d and trace energy flow from muscular effort through mechanical work to thermal energy. The DQB is updated: the mystery of the hot nut is now resolved, and a new question emerges about efficiency. The class energy-transformation model is updated with a dissipation arrow, advancing the cumulative storyline toward Lesson 7.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners are seated in groups of four. The teacher projects or reads	Teacher opens with: 'Last lesson we calculated that the mechanic	Think-Pair-Share anchored to the phenomenon. Sticky-note	Teacher circulates during silent writing and scans sticky notes for

📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Guess and Check, Work Backward (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	aloud a brief scenario: the jua kali apprentice at Gikomba notices the wheel nut feels warm after the mechanic made ten hard pushes with the short spanner — but the nut never moved. Each learner individually writes two predictions on a sticky note: (1) Where did the energy from the mechanic's ten pushes go if the nut never moved? (2) Would the nut get hotter if the mechanic pushed harder or just longer? Learners share predictions with their group partner before the teacher takes three or four responses from the class. Responses are posted on the Driving Question Board under the heading 'Our predictions about the hot nut.'	did zero joules of useful work with the short spanner because the nut did not move. Zero joules of work — yet the nut got hot. How is that possible? Take two minutes silently and write your prediction on the sticky note.' [WAIT TIME: 2 minutes silent individual writing — no talking allowed.] After writing, teacher says: 'Now share your idea with the person sitting next to you. You have one minute.' [WAIT TIME: 1 minute pair discussion.] Teacher then cold-calls three groups: 'Group near the window — what did you predict?' After each response teacher probes: 'What evidence from your own life makes you think that?' Teacher does NOT confirm or deny predictions at this stage, saying only: 'Interesting — let us keep that idea and test it.'	predictions posted on the DQB create a public record that learners will revisit after the investigation, enabling them to see which predictions were confirmed or revised.	two misconceptions to watch for: (1) 'The energy disappeared' — indicates a gap in energy conservation understanding from Lesson 2; (2) 'Friction creates new energy as heat' — indicates confusion about energy transformation vs. creation. Teacher mentally notes which groups hold these ideas to target questioning during the Explain phase.
Observe Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Guess and Check, Work Backward (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group of four receives: one smooth wooden block (approximately 10 cm x 8 cm x 4 cm, with a hook screwed into one end), one spring balance (0–10 N range), one 30 cm strip of sandpaper (coarse grade), one smooth ceramic tile or laminate offcut, a 30 cm ruler, and a data table printed on a worksheet. Groups follow a structured protocol. Step 1 — Smooth surface: Learner A places the block on the tile and pulls it with the spring balance at a slow, constant speed for exactly 20 cm. Learner B reads and records the spring balance while the block is moving (not at the moment it starts). Learner C measures the distance with the ruler. Learner D records in the table. The pull is repeated three times and results are averaged. Step 2 — Rough surface: The same protocol is	Teacher demonstrates the constant-speed pulling technique before groups begin: 'Pull slowly and steadily — we want the block moving at constant speed so that the spring balance reading equals the friction force. If you accelerate, the reading includes the force needed to speed it up and that is not what we want.' Teacher circulates and checks three specific things: (1) that learners are reading the spring balance while the block is in motion, not at rest; (2) that the sandpaper is flat and not bunching under the block; (3) that learners are recording three trials, not just one. When a group finishes early, teacher prompts: 'You have your numbers — now before you calculate, predict: which surface will give a larger value of work done against friction? Write that prediction in your table.' [WAIT TIME after this	Structured group investigation with assigned roles (Puller, Reader, Measurer, Recorder) ensures all learners are active. Three-trial averaging builds measurement literacy. The tactile warmth observation is a deliberate bridge between the quantitative data and the qualitative phenomenon of the hot wheel nut.	Teacher checks data tables for two quality indicators: friction force values that are consistent across three trials (variation of more than 1 N suggests the block was not pulled at constant speed — teacher re-demonstrates); and correct unit use (N for force, m for distance, J for work). Teacher asks any group that records 'J/m' or 'Nm' as the unit: 'Tell me what a joule is defined as — what two quantities multiply to give one joule?' to prompt self-correction.

	repeated with the sandpaper fixed to the desk beneath the block. Step 3 — Feel the surface: After ten consecutive pulls on the sandpaper, learners carefully press a fingertip on the sandpaper and on the block base for two seconds and record qualitative observations ('warm', 'slightly warm', 'no change'). Step 4 — Calculations: Each learner independently calculates $W_f = f_r \times d$ for both surfaces using the average friction force and d = 0.20 m, recording answers in joules.	prompt: 30 seconds for learners to write before teacher moves on.] After the feel-the-surface step, teacher asks the whole class: 'Hands up if your sandpaper felt warmer than the tile after ten pulls.' Teacher notes the show of hands publicly: 'Almost everyone — that is important evidence. What does it tell us?'		
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Guess and Check, Work Backward (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher draws a simple energy-flow diagram on the board with three boxes: [Mechanic's Muscles] → [Mechanical Work on Block] → [?]. Learners are asked to fill in the third box based on their investigation. Groups discuss for 90 seconds, then share. The teacher guides the class to agree: the third box is [Thermal Energy in surfaces]. The teacher then writes the formula $W_f = f_r \times d$ on the board and connects it to the class data: 'Using your average friction force on sandpaper and a distance of 0.20 m, what value of work done against friction did you get?' Two or three groups share their calculated values and the class compares them. The teacher then poses the central explanation question: 'The mechanic pushed the wheel nut ten times with the short spanner. The nut did not move, so displacement was zero and work done on the nut was zero joules. Yet the nut got hot. Use today's formula — $W_f = f_r \times d$ — to explain where the mechanic's energy went.' Learners write a two-sentence explanation in their notebooks, then two	Teacher uses a Socratic sequence after learners share their third-box answers. If a group says 'heat', teacher probes: 'Where exactly does the heat go — into which objects?' to push for precision (into both the block surface and the tile or sandpaper). If a group says 'the energy is lost', teacher says: 'Lost means it no longer exists — but our energy conservation principle from Lesson 2 says energy cannot be created or destroyed. So instead of lost, what word should we use?' [WAIT TIME: 10 seconds.] Teacher then writes the two-sentence explanation task: 'Write independently for two minutes — I want two sentences: one using $W_f = f_r \times d$ and one mentioning energy conservation.' [WAIT TIME: 2 minutes silent writing — teacher does not speak during this time.] After two volunteers share, teacher asks the class: 'Does this explanation match your original sticky-note prediction? Give me a thumbs up if it matched, sideways if partly, down if you were surprised.' Teacher notes the distribution and briefly validates surprises: 'Being	Energy-flow diagram with a missing box drives learners to construct the explanation before it is given. The two-sentence writing task forces individual consolidation. The thumbs metacognitive check connects back to predictions and models the scientific practice of revising ideas based on evidence.	Teacher collects two written explanation notebooks from each table (selected at random by asking for the notebook of the person born earliest in the year). These are scanned quickly for: correct use of $W_f = f_r \times d$ with units; the phrase 'thermal energy' or 'heat energy'; and an explicit mention that total energy is conserved. Notebooks with missing elements are returned with a single written prompt question before the next lesson.

	volunteers read theirs aloud. The class refines the explanation together. The teacher formalises: 'Work done against friction converts mechanical energy irreversibly into thermal energy. The total energy is still conserved — it has not disappeared — but it has changed into a form we cannot easily use again to tighten or loosen a nut.' The teacher connects to a local context: 'This is exactly why the brake pads on a Nairobi matatu become very hot going down the Limuru escarpment — the driver is doing work against friction deliberately to slow down, and all that mechanical energy becomes heat in the brakes.'	surprised is exactly how science works — our investigation changed our thinking.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Guess and Check, Work Backward (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the DQB and their sticky-note predictions posted at the start of the lesson. Each group retrieves their own sticky note and adds a second sticky note in a different colour answering: 'What did we actually find out about where the energy goes?' The teacher facilitates a three-minute whole-class review of how the DQB has evolved across lessons 1 to 4. A learner volunteer reads out the original DQB question from Lesson 2: 'If energy is conserved, why does the wheel nut get hot?' The class agrees this question is now answered and the teacher places a green checkmark sticker on it. However, the teacher poses a new unresolved question: 'We now know friction wastes some of the mechanic's work as heat. So when the mechanic uses the long spanner and successfully loosens the nut, is all of his work input going into actually loosening the nut — or is some still wasted	Teacher says: 'Look at your original prediction sticky note. In one sentence — write whether your prediction was correct, partly correct, or incorrect, and why. Be honest — scientists update their ideas when evidence changes them.' [WAIT TIME: 1 minute.] Teacher then facilitates the DQB review at a brisk pace, pointing to each sticky note cluster and asking: 'Is this question now answered? Partly answered? Still open?' Teacher models intellectual honesty by pointing to any prediction that was completely wrong and saying: 'This group predicted the energy disappeared — and they have now revised that to energy is transformed. That revision is not a mistake — that is exactly what scientists do.' Teacher then adds the new DQB question about efficiency, saying: 'This new question is a bridge to our next lessons on machines and efficiency. Write it in your notebook	DQB review makes the progression of understanding visible across lessons, reinforcing the storyline. The green-checkmark ritual gives learners a sense of intellectual accomplishment. Adding a new question models that resolving one mystery often generates deeper questions — a key scientific disposition.	Teacher observes whether learners can correctly identify the Lesson 2 DQB question as now answered. If any group says it is still unanswered, teacher asks: 'What did our investigation today tell us about where the heat comes from?' and uses the response to identify whether the Explain phase explanation was internalised or only surface-level recalled.

	as heat even when the nut moves?' This question is added to the DQB as a new yellow sticky note to be answered in Lessons 6 and 7.	as a question you now want to answer.'		
Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Guess and Check, Work Backward (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups return to the class energy-transformation model diagram that was started in Lesson 2 (KE and PE interchange) and updated in Lesson 3 (work done by applied force). Each group has a printed copy of the current model. They are given three minutes to add a new red arrow labelled 'Work done against friction' pointing away from the main energy flow pathway into a box labelled 'Thermal Energy (heat in surfaces)'. They also add the formula W_f = f_r x d next to the arrow. A note is added: 'This energy cannot be converted back to useful mechanical work — it is dissipated.' One group volunteer presents their updated model to the class. The teacher draws the agreed version on the board and photographs it for display. Learners copy the updated model into their notebooks. In the final two minutes, each learner independently writes a one-sentence answer to the lesson objective question: 'Why does the wheel nut in the Gikomba scenario feel warm after the mechanic pushes hard with the short spanner even though the nut never moved?'	Teacher circulates during model updating and checks that learners are placing the dissipation arrow correctly — branching off the main energy pathway, not replacing it. Common error to watch for: learners draw the thermal energy box at the end of the chain as if all energy becomes heat, rather than as a side branch showing partial dissipation. Teacher corrects this by asking: 'Is ALL the energy going to heat, or only the part that friction takes? Show me with the size of your arrow.' After the group presentation, teacher asks the whole class: 'Does this model now explain both mysteries from our original phenomenon — the mechanical advantage of the long spanner AND why the nut gets hot?' Learners respond and teacher says: 'We have answered the second mystery today. The first mystery — the mechanical advantage — is still sitting on our DQB. We will return to it in Lesson 6.' Teacher closes with: 'In your notebook, write the date and one sentence answering today's question about the warm nut. This is your exit evidence for today.'	Cumulative model building across lessons makes conceptual progression concrete and visual. The distinction between the main energy pathway and the dissipation side-branch is a key representational move that prepares learners for the efficiency calculation in Lesson 7. The final one-sentence exit response provides individual accountability and a formative data point for the teacher.	Teacher collects exit sentences as learners leave (or reads them over shoulders during the final two minutes). Target response should include: friction force, distance, thermal energy, and a statement that total energy is conserved. Any exit sentence that says only 'friction makes heat' without connecting to work (force x distance) indicates that the formula W_f = f_r x d has not yet been applied to the phenomenon — teacher notes these learners for a brief check-in at the start of Lesson 5.

D. TEACHER REFLECTION

After the lesson, consider: Did the tactile warmth observation during the Observe phase create genuine surprise or was it treated as routine? If learners were not surprised, the connection to the hot-nut phenomenon may need to be made more explicit at the start of Lesson 5. Did the energy-flow model updating feel mechanical (copying) or did learners propose where to place the dissipation arrow themselves? If mostly copying, consider giving groups a blank model in Lesson 7 to reconstruct from memory. Were the exit sentences using W_f = f_r x d with the phenomenon context, or were they generic definitions? Generic responses suggest the phenomenon

connection was lost during the Explain phase — consider opening Lesson 5 with a one-minute revisit of the Gikomba scenario before moving to power.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We pulled a wooden block across a smooth tile and a sandpaper surface using a spring balance and measured the friction force. We calculated the work done against friction using W_f = f_r x d. After ten pulls on sandpaper, the surface felt noticeably warmer than after pulls on the smooth tile.
What did I learn?	Work done against friction converts mechanical energy into thermal energy. The formula is W_f = f_r x d where f_r is the friction force in newtons and d is the distance slid in metres. Total energy is conserved — the thermal energy produced equals the mechanical energy input that went against friction. Rougher surfaces produce larger friction forces and therefore more thermal energy for the same distance.
How does this explain the phenomenon?	The wheel nut at Gikomba felt warm after the mechanic pushed hard with the short spanner because even though the nut did not move (zero useful work output), there was still microscopic relative movement and deformation at the contact surfaces between the spanner and nut, and between the nut and bolt thread, converting the mechanic's muscular energy into thermal energy through friction. This is why the nut gets hot without tightening or loosening — the energy was not lost but transformed irreversibly into heat.

LESSON 5 (80 minutes): Power: Definition, Calculation, and Everyday Applications	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to define power, calculate power output in real-world scenarios, and connect the concept of power to the rate of energy transfer in mechanical systems — anchored in the jua kali mechanic phenomenon.
Knowledge	Learners will define power as the rate of doing work (P = W/t), distinguish between power and energy, identify the SI unit of power (Watt), and explain how power relates to force and velocity (P = Fv).
Skills	Learners will calculate power output from given work and time data, compare power outputs across different scenarios (stair-climbing, Kipchoge's marathon, Mwea irrigation pump), and use Khan Academy interactive problems to consolidate numerical fluency.
Attitudes	Learners will appreciate that being more powerful does not mean being more efficient, and will value the role of physics in understanding real-world Kenyan mechanical and agricultural systems.
Key Inquiry Question	How is energy conserved and transformed in mechanical systems? How do simple machines make work easier without creating extra energy?
Purpose in Storyline	In Lessons 1–4, learners established what work is, how energy transforms, how mechanical advantage allows the jua kali mechanic's long spanner to multiply force, and how friction dissipates energy as heat in the wheel nut. Lesson 5 now asks: what if two mechanics loosen the same nut — one slowly, one quickly? Both do the same work, but something is different. That something is POWER. By calculating and comparing power outputs — in stair-climbing, in Eliud Kipchoge's marathon, and in a Mwea irrigation pump — learners build the critical bridge between work done and the rate at which it is done, setting up the final lessons on efficiency and machines.
Safety Notes	No physical hazards in this lesson. If learners physically climb stairs for the extension activity, ensure the staircase is clear of obstacles, learners wear appropriate footwear, and no learner with a known medical condition participates in timed climbing without teacher approval. All digital activities should be supervised and screen time managed responsibly.

B. LESSON OVERVIEW

Learners begin by predicting whether two people doing the same amount of work are equally 'powerful'. They then watch a YouTube stair-climbing demonstration video and analyse data comparing a student and a teacher climbing the same staircase at different speeds. Using the formula P = W/t, they calculate each person's power output and discover that power measures the RATE of working, not the total work done. The lesson pivots to two iconic Kenyan contexts: (1) Eliud Kipchoge's average power output during his sub-2-hour marathon attempt, and (2) the wattage rating of a pump used in the Mwea Irrigation Scheme in Kirinyaga County. Khan Academy interactive problems consolidate calculation skills. The lesson closes with a DQB revisit: 'So a more powerful machine does the same work faster — but is it more efficient?' — deliberately leaving efficiency as an unresolved productive question to anchor Lesson 6.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners are presented with a written scenario on the board:	Display the Kamau/Odhiambo scenario on the board or projector.	Predict-then-Revisit (Elicit Prior Knowledge): By publicly	Observable indicator: Can learners articulate a reason for their

📹 **VIDEO:** [AP Physics 1 Review of Charge and Circuits] Source: Khan Academy (English - US curriculum) [🔍 Search ARES for similar videos] 📖 **HTML:** [Raising a Product or Quotient to a Power (Flexbook)] Source: CK-12 [🔍 Search ARES for similar readings]	'Kamau and Mr. Odhiambo both carry the same box of tools (mass = 20 kg) up the same staircase at Gikomba market (height = 4 m). Kamau takes 8 seconds; Mr. Odhiambo takes 40 seconds. Who did more work? Who was more powerful?' Learners write individual predictions in their exercise books with reasons (2 minutes), then share with a partner (1 minute). A show-of-hands class poll is conducted for each option: (a) Kamau did more work, (b) Mr. Odhiambo did more work, (c) both did the same work. The same poll is repeated for 'more powerful'. Poll results are recorded on the board as a class tally to revisit at the end of the lesson.	Say verbatim: 'Before we calculate anything, I want your honest gut feeling. Write your prediction and your reason — even if you are not sure, write something. You have 2 minutes of silence.' [WAIT 2 minutes — do not prompt.] Then: 'Now turn to your desk partner and compare your predictions. Do you agree? Why or why not? 1 minute.' [WAIT 1 minute.] Conduct the show-of-hands poll. Cold-call 3 learners who chose DIFFERENT options: 'Amina, you said both did the same work — tell us your reasoning.' [WAIT 10 seconds.] 'Brian, you said Kamau did more work — what made you think that?' [WAIT 10 seconds.] Record all responses on the board under the heading 'OUR PREDICTIONS — revisit at end.' Do NOT correct predictions yet. Say: 'Let us find out what the physics actually says.'	committing to a prediction before instruction, learners activate their existing mental models about work and power. The class tally makes misconceptions visible — particularly the common conflation of 'working harder/faster' with 'doing more work'. This sets up productive cognitive dissonance that the observation phase will resolve.	prediction, even if incorrect? Listen specifically for learners who conflate speed with work done — this reveals the target misconception. Record on clipboard: how many learners (roughly) correctly predict that both do the same work but differ in power? This informs pacing of the explain phase.
Observe Phase 📹 **VIDEO:** [AP Physics 1 Review of Charge and Circuits] Source: Khan Academy (English - US curriculum) [🔍 Search ARES for similar videos] 📖 **HTML:** [Raising a Product or Quotient to a Power (Flexbook)] Source: CK-12 [🔍 Search ARES for similar readings]	Part A — YouTube Stair-Climbing Demo (15 min): The class watches a short YouTube clip (teacher pre-selected, e.g. a physics demonstration of two people climbing stairs at different speeds, with mass and height data shown). Learners are given a structured observation table to complete during the video: columns for Person, Mass (kg), Height climbed (m), Time taken (s), Work done (J) [to be calculated], Power (W) [to be calculated]. The video is paused at key moments for learners to record values. Part B — Data Set Analysis (10 min): Learners receive a printed or projected data table with THREE scenarios: (1) Student Achieng climbs school stairs: mass 50 kg, height 3 m, time 6 s. (2) Teacher Mr. Kariuki climbs same stairs:	Before playing the video, say: 'As you watch, your job is to be a data collector — like a scientist at a sports event. Fill in the observation table. I will pause the video so you can write.' Play the video, pausing at moments when data appears on screen. [WAIT 15 seconds at each pause for writing.] After the video, say: 'Now use the data you collected to calculate Work and Power for each person. Use W = mgh and P = W/t. Work in pairs. You have 8 minutes.' [Circulate — crouch to eye level with pairs, do NOT give answers, ask: 'What formula are you using? What does each letter stand for?'] After pair work, cold-call 3 pairs to share their Achieng, Kariuki, and Kipchoge calculations on the board. Say: 'Does everyone agree with these numbers? Thumbs up if	Data-Driven Pattern Recognition with Anchored Contexts: Learners do not just receive the formula — they extract patterns from real data. By juxtaposing Achieng (less mass, faster, less total work but possibly more power), Mr. Kariuki (more mass, slower), and Kipchoge (real-world elite athlete), learners see that power is fundamentally about rate. The Mwea pump problem bridges physics to Kenyan food security infrastructure, making the concept immediately relevant and motivating.	Observable indicators: (1) Are learners correctly applying W = mgh and P = W/t without prompting? (2) Do learners correctly identify that Achieng and Mr. Kariuki do different amounts of work (different masses) — catching the nuance that the scenario in Phase 1 used the SAME mass? (3) For the Mwea pump: can learners independently determine that mass = volume × density before applying P = W/t? Circulate and use a checklist: mark pairs as 'confident', 'developing', or 'needs support' for targeted intervention in Phase 3.

	mass 80 kg, height 3 m, time 15 s. (3) Eliud Kipchoge running his sub-2-hour marathon: average force ~ 800 N, average speed ~ 5.85 m/s. They work in pairs to calculate Work done (W = mgh for stair cases) and Power (P = W/t) for scenarios 1 and 2, and P = Fv for Kipchoge. A third scenario is projected as a 'Kenyan context extension': The Mwea Irrigation Scheme pump lifts 500 litres of water per minute to a height of 8 m. What is its power rating in kilowatts? (g = 10 N/kg; density of water = 1000 kg/m³).	yes, thumbs sideways if unsure.' Address any disagreements before moving on. Then say: 'Now here is a Kenyan challenge — the Mwea Irrigation pump. This one uses P = W/t but you will need to first find the mass of the water. Think about how to do that.' [WAIT 15 seconds.] Cold-call 1 learner to explain their approach before all pairs calculate.		
Explain Phase 📹 **VIDEO:** AP Physics 1 Review of Charge and Circuits Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Raising a Product or Quotient to a Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Formalising the Concept (8 min): Learners use their calculated results to construct a formal definition of power in their own words, then compare with the textbook/KICD definition. They record: Power = Work done ÷ Time taken; P = W/t; Unit: Watt (W) = Joule per second (J/s). They also derive P = Fv from P = W/t and W = Fd (since v = d/t). Part B — Khan Academy Interactive Problems (12 min): Learners access Khan Academy (offline via ARES platform) and complete the 'Power' practice exercise set. The set includes 5 problems of increasing difficulty: basic P = W/t, unit conversion (kW, MW), P = Fv application, and a multi-step problem. Learners work individually. Part C — Revisiting the Phenomenon (5 min): The teacher projects two questions: 'Q1: Our jua kali mechanic at Gikomba finally loosens the wheel nut with the long spanner. He does it in 2 seconds. His apprentice, using the same long spanner, loosens a second nut in 10 seconds. They did the same work.	After calculations are confirmed, say: 'Based on everything you just calculated, I want each of you to write — in your own words, not the textbook's words — what power means. You have 90 seconds of silence.' [WAIT 90 seconds.] Cold-call 4 learners to share their definitions. Write key phrases from student responses on the board. Then say: 'Now let us see how close you are to what physicists say.' Reveal the formal definition. Say: 'Open ARES — Khan Academy Power problems. Work through all 5 individually. If you finish early, check your Mwea pump answer from earlier. You have 12 minutes.' [Circulate — target the 'needs support' pairs identified in Phase 2. Spend 2–3 minutes with each struggling learner: 'Tell me what you know. What is the question asking for?'] After Khan Academy, project the Gikomba phenomenon questions. Say: 'In your groups of four — Discuss Q1 and Q2. You have 3 minutes. Then each group gives me ONE sentence for each question.' [WAIT 3 minutes.] Cold-	Construct-Before-Reveal (Generative Learning) + Phenomenon Re-anchoring: Learners write their own definition before seeing the formal one, which forces active synthesis rather than passive copying. Re-anchoring to the Gikomba mechanic in Q1 and Q2 ensures that the abstract formula (P = W/t) is always tethered to the physical reality that opened the unit. Connecting friction heating to power (rate of energy dissipation) deepens the link between Lessons 3–4 (energy dissipation) and Lesson 5 (rate of energy transfer).	Observable indicators: (1) Do learner-generated definitions include the word 'rate' or an equivalent idea ('how fast', 'per second', 'how quickly')? Definitions that only say 'how much work' reveal residual confusion. (2) Khan Academy completion rate and accuracy — ARES platform logs responses; teacher reviews at end of lesson. (3) For phenomenon Q1: can learners correctly state that the mechanic is more powerful AND explain it using P = W/t? (4) For Q2: do any learners spontaneously connect power to the rate of friction heating (thermal power)? This is an extension-level insight — celebrate it publicly if it emerges.

	Who was more powerful? Q2: Earlier in our unit, the nut got HOT from friction. Was that energy transfer powerful? What does 'power' mean for the friction heating process?' Learners discuss in groups of four and write a group response.	call 2 groups. Affirm correct reasoning; gently probe incomplete answers: 'You said the mechanic was more powerful — can you say that using numbers or the formula?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** AP Physics 1 Review of Charge and Circuits Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Raising a Product or Quotient to a Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class DQB (created in Lesson 1 and updated in each subsequent lesson) is displayed. Learners individually review the existing questions and sticky notes. Each learner writes on a new sticky note: one thing they can NOW answer on the DQB using today's learning about power. They also write one NEW question that today's lesson raised — especially focusing on the teacher's closing provocation: 'A more powerful machine does the same work faster — but is it more efficient?' Sticky notes are added to the DQB in two columns: 'NOW ANSWERED' and 'NEW QUESTIONS'. The class reads aloud three 'Now Answered' notes and three 'New Questions' notes selected by the teacher.	Direct learners to stand and face the DQB. Say: 'We have been building this board since Lesson 1. Look at the questions we wrote at the beginning. Which ones can you now answer — fully or partly — because of what you learned today about power?' [WAIT 20 seconds of reading time.] Distribute sticky notes. Say: 'Write one thing you can now answer. Be specific — use the word power, or the formula, or a number. Then write one new question today raised for you. You have 3 minutes.' [WAIT 3 minutes.] Collect and scan notes as learners post them. Select 3 'Now Answered' notes that best represent the class's understanding — read them aloud and say 'Does everyone agree this is answered? Thumbs up.' Select 3 'New Questions' notes. Read them aloud. Highlight any that connect to efficiency. Say: 'This question — IS a more powerful machine more efficient? — is exactly what we will investigate next lesson. Hold that thought.' Do NOT answer it. Leave it visibly posted on the DQB under 'Our Next Investigation'.	Driving Question Board as Cumulative Sense-Making Artefact: The DQB makes thinking visible across the entire unit. By distinguishing 'Now Answered' from 'New Questions', learners experience scientific inquiry as an iterative, growing process rather than a linear march through facts. The deliberate non-answer to the efficiency question creates productive intellectual tension — a hallmark of storyline teaching — that will motivate Lesson 6.	Observable indicators: (1) Quality of 'Now Answered' sticky notes — do they use precise language (power, Watts, rate, P = W/t) or vague language ('it is faster')? This reveals depth of conceptual uptake. (2) Quality of 'New Questions' — do learners distinguish power from efficiency, or do they conflate them? Conflation is the key misconception to address in Lesson 6. (3) Count how many learners' new questions specifically mention 'efficiency' — this shows readiness for the next lesson's concept.
Model Building Phase 📹 **VIDEO:** AP Physics 1 Review of Charge	Learners retrieve their cumulative unit model — a diagram/annotated sketch of the Gikomba jua kali mechanic scenario that they have revised in each lesson. In Lessons 1–4, they added: force arrows,	Say: 'Get out your unit model — the drawing of the mechanic that we have been updating every lesson. Today we add the power layer.' Display a teacher version of the model on the projector	Cumulative Model Revision (Progressive Modelling): The iterative model — revised in every lesson — externalises the learner's growing understanding as a single coherent artefact. Adding a 'RATE	Observable indicators: (1) Does the model correctly show power as a rate (J/s or W), not just a large force or a lot of work? (2) Is the power calculation on the model numerically correct? (3) Does the

and Circuits Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Raising a Product or Quotient to a Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	work calculations, energy transformation labels (kinetic → potential → thermal via friction), and mechanical advantage annotations. Today, they add a POWER LAYER to their model: (1) They annotate the mechanic's action with a power calculation: 'If the mechanic applies 200 N force over 0.3 m in 2 seconds, P = W/t = (200 × 0.3) / 2 = 30 W.' (2) They add a 'RATE arrow' — a new visual element showing that power is about how fast energy flows through the system, not just the total amount. (3) They add a thought bubble from the apprentice: 'The nut heated up over 30 seconds of struggling — the friction was dissipating energy at a low but continuous rate.' (4) Learners write one sentence at the bottom of their model: 'Power tells us HOW FAST work is done, not HOW MUCH work is done. The long spanner made the job easier (mechanical advantage), but it was the mechanic's speed that determined his power output.' Pairs share their updated models with an adjacent pair and give one piece of feedback.	showing Lessons 1–4 annotations. Say: 'You have 8 minutes to add the power annotations I described. Use the numbers from your calculations today. If you need the formula, it is on the board.' [WAIT 8 minutes — circulate, look specifically for whether learners are adding RATE language, not just numbers.] After 8 minutes: 'Turn to the pair next to you. Look at their model. Give them one piece of specific feedback: either something they explained really clearly, or something you think is missing or unclear. 2 minutes.' [WAIT 2 minutes.] Cold-call 2 learners to share their one-sentence summary at the bottom of their model. Say: 'Next lesson, we are going to ask: if the mechanic uses a more powerful tool, does he actually get MORE work out of the energy he puts in? That is the question of EFFICIENCY — and our model will need one more layer.' Collect 4–5 models to review overnight for formative feedback.	arrow' as a new visual vocabulary item embeds the distinction between work (a quantity) and power (a rate) in a spatial, memorable way. Peer feedback during model sharing activates metacognition — learners must evaluate another's scientific representation, not just their own. Collecting models allows the teacher to conduct formative assessment between lessons.	'Rate arrow' visually distinguish power from work in the diagram? (4) Does the apprentice's thought bubble reflect understanding that friction heating is also an energy transfer at a rate (thermal power), even if learners do not use that exact term? Collect 4–5 models and write one written comment per model returned next lesson.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions before Lesson 6:

1. POWER vs. WORK CONFUSION: During the predict phase, how many learners conflated 'working faster' with 'doing more work'? Did the stair-climbing observation phase successfully resolve this misconception? If more than 30% of learners still conflate the two concepts in their DQB sticky notes, plan a 5-minute re-teaching opener for Lesson 6 using the Kamau/Odhiambo scenario with explicit numerical comparison.

2. KIPCHOGE AND MWEA ENGAGEMENT: Did the Kipchoge marathon power calculation and the Mwea irrigation pump problem generate genuine engagement and discussion? If learners struggled with P = Fv (Kipchoge) or with finding mass from volume (Mwea), note which calculation step caused the breakdown — this informs how scaffolded Lesson 6's efficiency calculations should be.

3. DQB QUALITY: Review the 'New Questions' sticky notes. Are learners distinguishing power from efficiency, or using them interchangeably? If interchangeable use is widespread, begin Lesson 6 with a deliberate vocabulary contrast: 'Power = how fast. Efficiency = how much of what you put in comes out as useful work.'

4. MODEL PROGRESSION: From the 4–5 models collected, assess whether learners are successfully integrating each lesson's new concept into a coherent cumulative

model, or whether models are becoming cluttered and incoherent. If the latter, consider providing a structured model template for Lesson 6 to help learners organise their thinking spatially. 5. PERSONAL RELEVANCE: Did learners spontaneously make connections to other Kenyan power contexts beyond those offered (e.g., maize grinding mills in Nakuru, water pumps in Turkana, boda boda engine power)? Note these for use in Lesson 6's efficiency contexts — learner-generated examples drive deeper engagement.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Watched a stair-climbing YouTube demonstration showing a student (Achieng) and a teacher (Mr. Kariuki) climbing the same staircase at different speeds, and collected mass, height, and time data into an observation table. Reviewed data from Eliud Kipchoge's sub-2-hour marathon attempt and the Mwea Irrigation Scheme pump specifications.
What did I learn?	Power is the rate of doing work: P = W/t, measured in Watts (W = J/s). Doing the same work in less time requires more power. Power can also be calculated as P = Fv. A more powerful person or machine does not necessarily do more total work — they do the same work faster. The jua kali mechanic's speed of loosening the nut determines his power output; the long spanner determined his mechanical advantage. Friction heating the wheel nut is also an energy transfer happening at a rate.
How does this explain the phenomenon?	Two people doing the same amount of work are NOT equally powerful if they take different amounts of time. Power measures HOW FAST work is done, not HOW MUCH work is done. Eliud Kipchoge outputs roughly 1,000 W (1 kW) during peak marathon running — comparable to a household appliance. The Mwea irrigation pump's power rating can be calculated from how fast it moves water against gravity. A more powerful machine does the same work faster, but this does NOT mean it is more efficient — efficiency is about how much of the input energy becomes useful output energy, which is our next investigation.

LESSON 6 (80 minutes): Simple Machines: Types and Mechanical Advantage	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners identify the six types of simple machines in Kenyan contexts and calculate mechanical advantage (MA = Load ÷ Effort) using spring balances and constructed models, explaining why the long spanner handle at Gikomba market gives the mechanic a greater mechanical advantage.
Knowledge	Learners will know: (a) the six types of simple machines (lever, pulley, inclined plane, wheel and axle, screw, wedge); (b) how to calculate mechanical advantage using MA = Load ÷ Effort; (c) that simple machines do not create extra energy — they trade force for distance; (d) how each machine type relates to real Kenyan tools and workplaces.
Skills	Learners will be able to: (a) identify and classify simple machines from photographs and physical examples in Kenyan contexts; (b) measure load and effort using spring balances to calculate MA experimentally; (c) construct and investigate a simple lever model; (d) use data to explain why a longer spanner handle increases mechanical advantage; (e) represent machine systems using diagrams annotated with force arrows and distances.
Attitudes	Learners will appreciate: (a) how Kenyan jua kali workers intuitively apply mechanical advantage principles; (b) the value of scientific investigation in explaining everyday observations; (c) collaborative problem-solving when conducting experiments; (d) the ingenuity of informal-sector workers who maximise effort without formal physics training.
Key Inquiry Question	How do simple machines make work easier without creating extra energy?
Purpose in Storyline	This is Lesson 6 of 8 in the evidence-gathering sequence. Having established in earlier lessons that work = force × distance and that energy is conserved even when heat is produced, learners now directly investigate the mechanism behind the opening phenomenon: why does the long spanner extension give the mechanic a mechanical advantage? By measuring MA across six machine types — including a lever modelled on the spanner — learners gather concrete quantitative evidence to answer the driving question and prepare for Lesson 7's treatment of efficiency and energy dissipation.
Safety Notes	Ensure spring balances are not overloaded beyond their rated capacity. When constructing lever models, secure the fulcrum firmly so the plank does not slip and injure fingers. Masses and loads must be placed carefully on the lever; learners should stand clear when releasing heavy loads. Wedge and screw press demonstrations should be teacher-supervised — learners should not apply excessive force to screws or place fingers near the wedge edge.

B. LESSON OVERVIEW

In this 80-minute lesson, learners investigate all six types of simple machines using a structured gallery of photographs from Kenyan workplaces (a jua kali workshop lever, a Kisumu harbour pulley, a Kericho tea farm inclined ramp, a borehole hand-pump wheel and axle, a sugarcane screw press, and a wedge-shaped panga/machete). They begin by predicting which machine type gives the greatest mechanical advantage, then rotate through six lab stations where they use spring balances and physical models to measure MA = Load ÷ Effort. Through structured sensemaking, they discover that every machine trades force for distance — no machine creates extra energy — and that the long spanner extension in the Gikomba phenomenon is a classic first-class lever whose MA increases directly with arm length. The lesson advances the driving question by providing quantitative evidence that mechanical advantage is real, measurable, and bounded by the conservation of energy established in previous lessons.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Spout's reversing circuit and final assembly Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Simple Harmonic Motion (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a single projected image: the Gikomba jua kali mechanic using the long-pipe spanner extension alongside a side-by-side photograph of five Kenyan tools — a wheelbarrow (lever), a Kisumu harbour block-and-tackle pulley, a Kericho tea-farm loading ramp, a borehole hand-pump, and a panga machete. Each learner silently writes in their exercise book: (1) Which of these tools do you think gives the GREATEST mechanical advantage — i.e., lets you lift or move the heaviest load with the least effort? Rank them 1–5. (2) Predict the MA value for the spanner extension: if the mechanic pushes with 50 N of effort, how many newtons of force do you think acts on the nut? (3) Revisit the driving question on the board — 'Why can the mechanic loosen the nut effortlessly with the long spanner?' — and write one sentence predicting what determines how much easier a machine makes the job. Learners share predictions with a shoulder partner for 2 minutes, then three pairs share aloud with the class. Predictions are not corrected — they are recorded on a 'Prediction Wall' sticky-note section of the Driving Question Board for revisiting at the end of the lesson.	Display the dual photograph and say: 'Before we touch any equipment today, I want your honest prediction — not a guess based on what you think I want to hear, but your real thinking right now.' Allow 4 minutes of silent individual writing. Circulate and note 2–3 interesting or divergent predictions to use during cold-calling. After partner sharing, cold-call exactly 3 pairs using a name stick: 'Amina and Brian — what did you rank as the highest MA machine and why?' After each pair responds, ask: 'Does anyone have a DIFFERENT ranking? Raise your hand.' Allow 10 seconds WAIT TIME before calling on another learner. Record all predictions (without evaluating them) on sticky notes and place them on the DQB Prediction Wall. Close by saying: 'Keep these predictions in mind — by the end of today's stations, you will have data to test every one of them.'	Prediction with Public Commitment — By writing and sharing ranked predictions before any instruction, learners activate prior knowledge and surface misconceptions (e.g., 'bigger machine = more MA'). Public posting on the DQB creates accountability and cognitive dissonance when data contradicts predictions, which drives deeper engagement during the observation phase.	Observable indicator: Every learner has a written ranking and a predicted MA numerical value in their exercise book before station rotation begins. Teacher scans books during circulation; learners who have written only qualitative statements (e.g., 'the pulley is strongest') are prompted: 'Can you give me a number? If effort is 50 N, what is your predicted load force?' This reveals whether learners understand MA as a ratio.
Observe Phase 📹 **VIDEO:** Spout's reversing circuit and final assembly	Learners rotate through six lab stations in groups of four (approximately 8 minutes per station with 1 minute transition). Each station has a photograph card showing the Kenyan context,	Before rotation begins, say: 'You have 8 minutes at each station — that is not a lot of time, so read the instruction card FIRST before touching the apparatus. Every person in the group must read the	Structured Station Rotation with Comparative Data Table — Rotating through physical models of all six machine types within a single lesson gives learners direct sensory and quantitative	Observable indicator: Each learner's exercise book contains at least three rows of data per station (load, effort, MA) and a completed Comparative Data Table. Teacher checks: (a) Are MA values greater

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Simple Harmonic Motion (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	a physical model or apparatus, a spring balance, a known hanging mass (load), and a data-recording table printed on a card. Station 1 — LEVER (Jua Kali Spanner Model): A 1-metre wooden plank balanced on a triangular fulcrum (pivot block). A 500 g mass hangs as the load. Learners place effort at three different distances from the fulcrum (25 cm, 50 cm, 75 cm) and read the effort spring balance each time. They record Load (N), Effort (N), and calculate MA = Load ÷ Effort for each arm length. Photograph shows the Gikomba mechanic with the pipe extension. Station 2 — PULLEY (Kisumu Harbour Block and Tackle): A single fixed pulley and a double movable pulley system set up on a clamp stand. Learners lift a 1 kg mass using each and record the effort force from the spring balance. They calculate MA for each pulley arrangement and count the number of rope segments supporting the load. Station 3 — INCLINED PLANE (Kericho Tea Farm Ramp): A wooden board propped at three different angles (low, medium, steep) against a stack of books. A 500 g trolley (or block with wheels) is pulled up each incline while the spring balance reads effort. Learners record ramp length, height, effort, and calculate MA = Load ÷ Effort. They also calculate the ratio of ramp length to height. Station 4 — WHEEL AND AXLE	spring balance at least once. Assign roles before you start: one reader, one recorder, one measurer, one timekeeper.' During rotation, circulate continuously. At Station 1 (Lever), pause and ask: 'What happens to your effort reading as you move further from the fulcrum? Tell me in one word.' Allow 10 seconds WAIT TIME. At Station 3 (Inclined Plane), ask: 'You are getting a higher MA at the shallow angle — but you are pulling the trolley a longer distance. Does that remind you of anything from our earlier lessons on work?' Cold-call one learner by name. At Station 2 (Pulley), prompt: 'Count the rope segments holding the load — what do you notice about that number and your calculated MA?' Allow 15 seconds WAIT TIME before accepting an answer. With 2 minutes remaining at each station, say: 'If you have not yet calculated MA, do that now — do not move until you have a number.' At the end of rotations, say: 'Back in your home groups, complete your Comparative Data Table. You have 5 minutes. Be ready to share the machine with the highest and the lowest MA your group measured.'	experience with each class of simple machine. The Comparative Data Table forces side-by-side synthesis: learners cannot avoid noticing that MA values differ across machines and that some patterns (more rope segments = higher MA; longer ramp = higher MA; longer spanner arm = higher MA) repeat across apparently different machines. This pattern recognition is the conceptual foundation for the Explain phase.	than 1 for most machines? (b) Has the learner calculated MA as Load ÷ Effort, not the reverse? (c) At Station 1, does the MA increase as the effort arm lengthens? A common error — using effort ÷ load — is flagged immediately during circulation with the prompt: 'Which is bigger in a useful machine — the load or the effort? So which goes on top in the ratio?'

	(Borehole Hand-Pump): A spool-and-axle model (a large cotton reel with a pencil as axle and a handle rod threaded through). Learners hang a 200 g mass on the axle string and turn the handle, reading effort from a spring balance attached to the rim. They measure the wheel radius and axle radius and calculate MA = wheel radius ÷ axle radius and verify with Load ÷ Effort. Station 5 — SCREW (Sugarcane Press): A large bolt/screw (M12 or M16) with a known pitch (distance per revolution). Learners use a torque wrench or a bar handle to apply effort and lift a known load via the screw mechanism. They record pitch, effort arm, and estimate MA = ($2\pi \times$ effort arm) ÷ pitch. The photograph shows the Kisumu sugarcane screw press. Station 6 — WEDGE (Panga/Machete): A wooden wedge (door-stop shape) with known length and maximum thickness. Learners push the wedge under a heavy book stack using a measured force (spring balance) and observe the upward displacement. They calculate MA = length of wedge ÷ thickness at widest point and record. Photograph shows a panga machete used on a Rift Valley farm. Each group records all data in a shared Group Data Sheet and individually in their exercise books. After all stations, groups return to their home seats and complete a Comparative Data Table showing			

	all six machines side by side.			
Explain Phase 📹 **VIDEO:** [Spout's reversing circuit and final assembly] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Simple Harmonic Motion (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners remain in home groups. The teacher projects a blank six-row summary table (one row per machine type) with columns: Machine Type | Kenyan Example | Formula for MA | Measured MA | What is traded for the force gain? Learners spend 6 minutes completing this collaboratively using their station data. Groups then participate in a Jigsaw Share-Out: one member from each group moves to a new 'expert group' where each person represents a different station and teaches the others about their machine's MA and what is traded. Expert groups then disband and learners return to home groups to fill any gaps. The teacher then leads a whole-class Claim–Evidence–Reasoning (CER) discussion anchored to the phenomenon: 'Claim: Simple machines make work easier by increasing force — but they do NOT create extra energy. Evidence: Use your data. Reasoning: Connect to Work = Force × Distance from Lesson 3.' Learners individually write a CER paragraph in their exercise books answering: 'Why could the Gikomba mechanic loosen the nut with the long spanner? Use the words mechanical advantage, load, effort, distance, and work done in your answer.' They also annotate a diagram of a lever (drawn in their books) with: fulcrum position, effort arm length, load arm length, effort force arrow, load force arrow, and the calculated MA.	After the Jigsaw, project the phenomenon photograph again and say: 'The mechanic's pipe extension makes the spanner arm about 4 times longer. Using your Station 1 data, what MA do you predict that gives him?' Allow 15 seconds WAIT TIME. Cold-call two learners with different answers and ask: 'Walk me through your calculation.' After both respond, say: 'Who agrees with the first answer? Who agrees with the second? Hold up your data card.' Resolve the discrepancy using Station 1 data projected on screen. Then ask the critical conceptual question: 'Here is the puzzle — if the mechanic pushes with only 50 N of effort but gets 200 N on the nut, where did those extra 150 N come from? Did the machine CREATE energy?' Allow 20 seconds WAIT TIME — this is intentional productive struggle. Cold-call three learners. After responses, guide: 'Look at the distance your effort hand travels compared to the distance the nut moves. What do you notice?' Allow 10 seconds. Then say: 'This is the key insight of today's lesson — say it with me: Simple machines trade DISTANCE for FORCE. Work in equals work out — ignoring friction for now.' During CER writing, circulate and stamp or initial books where the reasoning explicitly mentions Work = F × d. Say to learners who have not included distance: 'Your claim and evidence are strong — now tell me WHY. What does work =	Jigsaw followed by Claim–Evidence–Reasoning (CER) Writing — The Jigsaw ensures every learner becomes a temporary 'expert' on one machine and must articulate its MA clearly enough to teach peers, deepening encoding. The CER writing then demands that learners synthesise all six machines into a single coherent principle (machines trade distance for force; energy is conserved) and apply it directly to the anchoring phenomenon. This dual strategy — peer teaching then independent writing — addresses both collaborative and individual sense-making.	Observable indicator: The CER paragraph must contain: (a) a specific MA value calculated from Station 1 data, (b) reference to effort arm length vs. load arm length, (c) an explicit statement that work input = work output (or that energy is conserved), and (d) connection to the Gikomba mechanic phenomenon. Teacher uses a 4-point checklist while circulating: 4 = all four elements; 3 = three elements; 2 = two elements; 1 = only a claim with no evidence or reasoning. Learners scoring 1 or 2 are invited to a 3-minute teacher-led small group during the DQB phase.

	Finally, learners revisit the Prediction Wall on the DQB and place a tick or cross next to their earlier predictions, correcting any that were wrong with a brief written justification.	force × distance have to do with this?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Spout's reversing circuit and final assembly] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Simple Harmonic Motion (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes. On the GREEN note they write one piece of evidence from today's lesson that helps answer the driving question ('Why can the mechanic loosen the nut effortlessly with the long spanner?'). On the ORANGE note they write one new question today's lesson has raised — for example: 'If MA = Load ÷ Effort, why isn't the MA infinite if I make the spanner infinitely long?' or 'We assumed work in = work out — but the nut gets hot, so is that actually true?' The class gathers at the DQB (a large poster or section of the classroom wall organised into: What We Know | What We Wonder | Evidence Gathered | Questions Remaining). Learners place their sticky notes in the appropriate sections. The teacher reads aloud three green notes and three orange notes, selected to highlight the strongest evidence and the most generative new questions. The class votes (thumbs up/down) on whether each green note qualifies as genuine evidence or is still just a claim needing more data. The teacher then draws a red arrow on the DQB from the 'Why does the nut get hot?' question	Say: 'Take two minutes — silent, individual — to write your green and orange sticky notes. I will be walking around and I should see everyone writing.' Circulate and prompt any learner with a blank note: 'What is ONE number from your data table that surprised you today? Write that as evidence.' After posting, read aloud the selected notes without attributing them to specific learners to protect psychological safety. When reading an orange 'wonder' question, pause and ask the class: 'Has anyone already gathered evidence that begins to answer this? Raise your hand.' Allow 10 seconds WAIT TIME before taking responses. Close by pointing to the full DQB and saying: 'Look at how much evidence we have now compared to Lesson 1. We are building a case. Two lessons remain — what do we still need to figure out to fully answer our driving question?'	Driving Question Board Curation — The DQB is not a passive display; it is a living record of the class's growing epistemic understanding. By distinguishing evidence (green) from new questions (orange), learners practise the science practice of constructing explanations and identifying gaps. The public vote on whether a claim qualifies as evidence develops scientific argumentation skills and reinforces that evidence must be data-based, not opinion-based.	Observable indicator: Green sticky notes contain a specific measurement or calculated MA value (e.g., 'When the effort arm was 75 cm, our lever gave MA = 4.2') rather than a vague statement ('levers are useful'). If a sticky note is a claim without data, the teacher returns it to the learner and says: 'This is a good claim — now add the number from your data table that proves it.' Orange notes are assessed for generativity: do they point toward the remaining lesson topics (efficiency, energy dissipation) or are they already answered by today's lesson?

	(posted in Lesson 2) to a new orange sticky note reading: 'We now know MA explains the force gain — but we still cannot fully explain the heat. That is Lesson 7.' This explicitly advances the storyline and maintains productive tension.			
Model Building Phase 📹 **VIDEO:** Spout's reversing circuit and final assembly Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Simple Harmonic Motion (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the cumulative model they have been building since Lesson 1 — a large annotated diagram in their exercise books (or on A3 paper) that represents the Gikomba mechanic's spanner system and traces energy and forces through it. In this lesson, they make the following specific additions to their model: 1. Add a LEVER DIAGRAM overlay on the spanner system: label fulcrum (nut/bolt interface), effort arm (length of spanner + pipe extension), load arm (short side of nut engagement), effort force arrow (mechanic's push), load force arrow (torque on nut). Annotate with MA = Load ÷ Effort and the measured value from Station 1. 2. Add a MACHINE TYPE KEY in the corner of their model: draw a small icon for each of the six simple machines, name the Kenyan example, and write the MA formula beside each. 3. Add a TRADE-OFF ANNOTATION: draw a two-column comparison — 'With short spanner: high effort, short distance' vs. 'With long spanner: low effort, long distance' — with	Say: 'Open your cumulative model from Lesson 1. You are going to add four things — I will give you the list, and you have 10 minutes. This is individual work — your model must reflect YOUR understanding.' Project the four additions as a numbered list. Circulate and look specifically for: (a) correct placement of the fulcrum on the lever diagram (it must be at the nut, not at the mechanic's hand); (b) MA value greater than 1 (if a learner has MA < 1 their effort arm and load arm may be swapped); (c) Work = F × d values that are equal on both sides of the trade-off column. When a learner has a correct lever diagram, ask: 'Now point to where on this diagram you would find the answer to why the apprentice's nut got hot.' Allow 15 seconds WAIT TIME. This connects to the unresolved heat question and previews Lesson 7. After partner comparison, cold-call two pairs to share differences: 'Tell me one thing your partner's model showed that yours did not — and whether you think they are right.' Close the lesson by saying: 'Your model is almost complete. By the end of Lesson 8, you will be able to draw and explain the full energy story — from the mechanic's muscles to the turning nut to the heat in the bolt — using every concept we have studied.'	Cumulative Model Revision with Annotated Lever Diagram — Requiring learners to physically add today's content to a model they have been building since Lesson 1 makes conceptual growth visible and forces integration of new knowledge (mechanical advantage, six machine types) with prior knowledge (work = F × d, energy conservation). The 'Question Flag' technique — marking an unresolved question directly on the model — maintains productive tension and ensures the heat/efficiency storyline is not dropped as new content accumulates.	Observable indicator: The cumulative model contains a correctly labelled lever diagram with MA value, a six-machine key with Kenyan examples, a trade-off annotation showing equal work on both sides, and a question flag at the friction point. The teacher photographs 4–5 models at different quality levels using a tablet or phone and uses them as anonymous comparison exemplars at the start of Lesson 7 to set expectations. The key misconception to flag: learners who draw the fulcrum at the middle of the spanner (rather than at the nut) have a fundamental misunderstanding of how a torque wrench functions as a lever — this must be corrected before Lesson 7.

	Work = F × d calculated for both sides to show they are equal (ideal, no friction). 4. Add a QUESTION FLAG: a small flag drawn on the model at the point where friction occurs (nut-bolt interface) with the text: 'Heat produced here — why? → Lesson 7.' Learners compare their updated model with their partner's and identify one element they have represented differently, discussing which representation is more accurate. Groups share one model addition with the class via a document camera or by holding up the A3 sheet.			

D. TEACHER REFLECTION

After the lesson, reflect on the following questions to inform Lesson 7 planning:

1. STATION DATA QUALITY: Did all six groups obtain MA values greater than 1.0 for most machines? If groups at Station 1 (Lever) did not show a clear increase in MA as the effort arm lengthened, was this due to friction in the plank setup, misreading of the spring balance, or conceptual confusion about which end is load vs. effort? Plan a 3-minute data correction discussion at the start of Lesson 7 if needed.

2. CER PARAGRAPH GAPS: How many learners achieved a score of 3 or 4 on the CER checklist? If fewer than 70% reached 3+, the class has not yet securely connected Work = F × d (Lesson 3) to the concept of MA trade-offs. Lesson 7's efficiency discussion depends on this link — consider a brief worked example at the start of Lesson 7 before introducing efficiency.

3. DQB ORANGE NOTES: Did learners' new questions cluster around efficiency and heat (pointing naturally toward Lesson 7), or did they reveal unresolved confusion about the six machine types themselves? If the latter, a quick 5-minute classification recap is needed at the start of Lesson 7.

4. EQUITY OBSERVATION: Were learners from different genders and ability levels equally active at the physical stations? Jua kali and farm contexts may resonate more strongly with learners from rural or informal-sector family backgrounds — did urban learners engage as fully? Consider assigning station leadership roles explicitly in Lesson 7 to ensure equitable participation.

5. MODEL BUILDING: Were the lever diagrams correctly oriented (fulcrum at the nut, not at the hand)? This misconception, if uncorrected, will cause errors in Lesson 7 when learners calculate efficiency using actual vs. ideal MA. Flag any persistent misplacements and address them using the Lesson 7 opening cold-call.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners rotated through six lab stations representing Kenyan simple machines — a jua kali lever, a Kisumu harbour pulley, a Kericho tea farm inclined ramp, a borehole hand-pump wheel and axle, a sugarcane screw press, and a panga wedge. At each

	station, they used spring balances and physical models to measure the effort required to move a known load, recording data in a Comparative Data Table.
What did I learn?	Learners calculated Mechanical Advantage (MA = Load ÷ Effort) for all six machine types and discovered that every machine with MA > 1 requires the effort to move through a greater distance than the load — confirming that Work In = Work Out (ideally) and that no machine creates extra energy. They connected this directly to the Gikomba phenomenon: the long spanner extension increases the lever's effort arm, raising MA and reducing the effort needed to loosen the nut.
How does this explain the phenomenon?	Using Claim–Evidence–Reasoning writing and a Jigsaw discussion, learners constructed the explanation that simple machines redistribute force and distance — trading one for the other while conserving total work done. They annotated cumulative models with a labelled lever diagram of the spanner system (fulcrum at the nut, effort arm = spanner + pipe extension, MA from measured data) and identified the friction point at the nut-bolt interface as the site of heat production — flagging this as the unresolved question for Lesson 7 on efficiency and energy dissipation.

LESSON 7 (80 minutes): Velocity Ratio, Efficiency, and the Cost of Friction	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to derive the Velocity Ratio (VR) of simple machines from distance measurements, calculate mechanical efficiency using Efficiency = (MA/VR) × 100%, and explain why no real machine achieves 100% efficiency due to energy dissipation through friction.
Knowledge	Learners will know that: (1) Velocity Ratio (VR) is the ratio of the distance moved by the effort to the distance moved by the load; (2) Efficiency = (MA/VR) × 100% and is always less than 100% in real machines; (3) the difference between useful work output and total work input is accounted for by energy lost to friction as heat; (4) machines redistribute force and distance but cannot create energy.
Skills	Learners will be able to: (a) measure effort distance and load distance on a simple machine to derive VR; (b) calculate efficiency given MA and VR values; (c) analyse and interpret efficiency data to identify friction as the primary energy loss mechanism; (d) update a cumulative explanatory model linking MA, VR, efficiency, and heat dissipation.
Attitudes	Learners will appreciate that understanding energy losses in machines has practical value for jua kali mechanics, engineers, and everyday Kenyans who depend on tools and machines for their livelihoods.
Key Inquiry Question	Why does a real machine always give back less useful work than we put in — and where exactly does the missing energy go?
Purpose in Storyline	This is Lesson 7 of 8 in the evidence-gathering sequence. Having established Mechanical Advantage (MA) in Lesson 6, learners now derive Velocity Ratio from their own distance measurements, compute efficiency, and confront the mathematical proof that no machine can be 100% efficient. This lesson closes the conceptual loop opened in Lesson 1 — the warm wheel nut is no longer mysterious; it is quantified energy loss. Learners update the class cumulative model to show that the jua kali mechanic's long spanner gains mechanical advantage but always sacrifices some energy to friction heat, answering the second strand of the driving question.
Safety Notes	When using lever setups with hanging masses, ensure retort stands are clamped securely to the bench. Do not exceed the load limits of spring balances. Learners should keep feet clear of falling masses. If a pulley and string system is used, tie string knots carefully and check before loading. Remind learners that real spanners and pipes should never be struck — demonstration is observational only.

B. LESSON OVERVIEW

Learners begin by revisiting their Driving Question Board (DQB) entries from Lessons 1–6, spotting the unanswered question: 'We know MA — but how far does the effort move compared to the load, and why can't we get all our work back?' They then measure effort distance and load distance on their lever and/or pulley models to derive Velocity Ratio (VR). Using their previously calculated MA values, they compute Efficiency = (MA/VR) × 100%, discover that all results fall below 100%, and reason — using evidence from the warm-nut phenomenon — that friction converts the 'missing' work into heat. A gallery walk of efficiency results from different machines (lever, pulley, inclined plane) reveals a pattern: higher friction → lower efficiency. Learners close by updating the cumulative class model diagram and recording a final DQB entry that directly answers the driving question's second strand.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown two scenarios	Display the two scenarios as a	Prediction Journaling with Think-	Observable indicator: Every

📹 **VIDEO:** Automotive mechanic: What I do and how much I make Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Angular Velocity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	on the board: (A) A jua kali mechanic in Gikomba uses a long-pipe spanner extension (MA = 4) to loosen a wheel nut. His effort hand moves 40 cm. (B) An apprentice uses a short spanner (MA = 2) and his effort hand also moves 40 cm. Learners are asked to predict, in writing in their notebooks: 'How far does the wheel nut (load) move in each case? If both mechanics push with the same force, who does more work on the nut — or is it equal? Where does any extra energy go?' Learners share predictions with a partner (Think-Pair), then three pairs are cold-called to share with the class. Predictions are pinned to the DQB as 'Our current thinking — Lesson 7 start.'	labelled diagram (sketch on board or projected image). Say: 'Before we calculate anything, I want your best scientific guess. Write it in your notebook — full sentences, not just numbers. You have 90 seconds.' [Allow 90 seconds of silent writing.] Then say: 'Share your prediction with your partner. Do you agree? Why or why not? 30 seconds.' [Allow 30 seconds.] Cold-call exactly 3 pairs: 'Pair at the window bench — what did you predict for Scenario A?' [WAIT 12 seconds after each cold-call before prompting.] After three pairs have shared, say: 'Notice we have disagreement — some say equal work, some say different. That tension is exactly what today's evidence will resolve. Let's hold those predictions on the DQB.'	Pair-Share. Purpose: activating prior knowledge about work (W = Fd) and creating cognitive dissonance — learners intuitively feel effort should equal output but cannot yet reconcile friction losses. This creates the need-to-know that drives the Observe phase.	learner has written at least two sentences predicting load distance and energy outcome before discussion begins. Teacher circulates during the 90-second window and notes (without correcting) whether learners invoke W = Fd, mention heat, or focus only on force. These patterns guide which misconceptions to address in Phase 3.
Observe Phase 📹 **VIDEO:** Automotive mechanic: What I do and how much I make Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Angular Velocity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Activity 1 — Measuring Velocity Ratio on a Lever (25 min): In groups of 4, learners set up a Class 1 lever (ruler or wooden beam pivoted on a pencil fulcrum, or a pre-assembled lever from the lab). They hang a 500 g load at a marked point on one side and use a spring balance as the effort on the other side. Using a ruler, they measure: (a) the distance the effort moves (effort distance, dE) when the load moves a fixed distance (load distance, dL = 5 cm). They record dE and dL, then calculate VR = dE / dL. They repeat for two different fulcrum positions, recording all data in a table. Activity 2 — Efficiency Calculation (10 min): Learners retrieve their MA values from Lesson 6 lab notebooks (or the teacher provides a reference sheet). They calculate Efficiency (%) = (MA / VR) × 100% for each	Distribute the lever setup to each group and say: 'Your job in the next 25 minutes is to become measurement scientists. Every centimetre matters. dE is how far your effort hand moves; dL is how far the load moves. Measure both with your ruler and record in your table.' Circulate every 4 minutes. At the 10-minute mark, pause all groups: 'Freeze. Hold up your VR values on your whiteboard slates [or call out]. Are they all the same? Should they be? Discuss for 30 seconds with your group.' [WAIT — allow 30 seconds of group talk.] Cold-call 2 groups to share VR values and fulcrum positions. Then say: 'Now open Lesson 6 notes and get your MA. Calculate efficiency. I want everyone to have a number before I count down from 5 — 5, 4, 3, 2, 1. Pens down.' For the video clip, say: 'Watch carefully — I will pause at the 1-	Structured Data Collection with Iterative Variation. Learners measure VR at two different fulcrum positions, generating a small dataset that shows VR changes with machine configuration — building the understanding that VR is a design property, not a fixed constant. The video cross-check introduces a second machine type (pulley/posho mill), broadening the evidence base beyond the classroom lever. Both activities tie directly back to the phenomenon: the jua kali mechanic's pipe extension changes the effort distance, which changes VR.	Observable indicator: Each group's data table has at least two complete rows (dE, dL, VR, MA, Efficiency) with units recorded. Teacher checks for the critical error of inverting VR = dL/dE (should be dE/dL) — if spotted in more than one group, pause the class for a 2-minute clarification before proceeding. Efficiency values above 100% signal a measurement or MA error and must be flagged immediately.

	fulcrum position. Learners record whether efficiency is above, at, or below 100%. Activity 3 — Video Clip Cross-check (5 min): The teacher plays a 2-minute clip from LabXchange (Energy and Work Modules) or a locally sourced video showing a pulley system lifting a sack of maize at a posho mill near Eldoret. Learners note the effort distance and load distance quoted in the clip and calculate VR and efficiency independently in their notebooks, then compare with a partner.	minute mark and ask you to tell me the effort and load distances. Do not write anything yet, just observe.' Pause, cold-call 2 learners for values, then say: 'Now calculate in your notebooks. 90 seconds.' [WAIT 90 seconds.] 'Compare with your partner — same answer? If not, find the arithmetic error together.'		
Explain Phase 📹 **VIDEO:** [Automotive mechanic: What I do and how much I make] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Angular Velocity (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Gallery Walk — Efficiency Across Machines (15 min): Each group posts their data table and efficiency calculation on the wall. Groups also receive a pre-printed card showing efficiency data for three other machines: a single fixed pulley (η ≈ 85%), a wheelbarrow (η ≈ 70%), and a screw jack used to lift a matatu (η ≈ 45%). Learners walk the gallery in silence for 5 minutes, annotating sticky notes with: (a) one pattern they notice, (b) one question they now have. Groups then return to seats for a 10-minute whole-class discussion anchored by the teacher. Conceptual Consolidation (10 min): Teacher leads a structured explanation using the board: 'Work input = Work output (useful) + Work lost to friction. So Efficiency < 100% always.' Learners complete a gap-fill in their notebooks: 'In the jua kali mechanic's spanner system, the effort moves ___ cm and the nut moves ___ cm, giving VR = ___. MA = ___. Efficiency = ___%. The missing ___% of energy is converted to ___ (heat) by ___.' Learners connect this directly to	During the Gallery Walk, circulate and listen — do not answer questions yet. After 5 minutes of silent annotation, call learners back: 'What pattern did everyone see? Someone from this side of the room — what did you notice?' [WAIT 12 seconds.] Cold-call 3 learners from different groups. Probe each with: 'Why do you think the screw jack is so much less efficient than the lever — what is happening physically between the screw threads?' [WAIT 10 seconds after each probe.] Then say: 'Let's put this in the language of physics.' Write on board: Efficiency (%) = (MA / VR) × 100% = (Useful Work Out / Total Work In) × 100%. Say: 'These two expressions are the same thing — convince me they are equivalent. Discuss with your partner for 60 seconds.' [WAIT 60 seconds.] Cold-call 1 pair to explain. Then say: 'The energy doesn't disappear — it moves. It moves into heat. We saw that heat in Lesson 1 — the warm wheel nut. Today we can finally calculate exactly how much energy the friction stole.' Complete the gap-fill as a class by calling on	Gallery Walk with Annotated Sticky Notes followed by Structured Whole-Class Discussion. The gallery walk externalises group thinking, allowing learners to see the efficiency pattern across multiple machines before it is named. The gap-fill bridges informal observation (warm nut) with formal quantitative reasoning (efficiency percentage), completing the explanatory arc of the driving question's second strand.	Observable indicator: During the gallery walk, every learner has written at least one pattern and one question on their sticky notes before the discussion begins. During the gap-fill, teacher collects 4 notebooks at random (one from each bench row) and checks: correct VR direction, efficiency below 100%, and the correct identification of 'heat' and 'friction' as the destination of lost energy. Misconception flag: if learners write 'energy is destroyed' rather than 'converted to heat,' this must be corrected immediately using the Law of Conservation of Energy.

	the warm-nut observation from Lesson 1.	individual learners for each blank.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Automotive mechanic: What I do and how much I make] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Angular Velocity (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board. The teacher displays the original driving question: 'Why can a jua kali mechanic in Nairobi loosen a wheel nut effortlessly with a long spanner when a short one fails — and where does the energy go when friction makes the nut hot?' Each learner writes a DQB update card individually (index card or sticky note) responding to: 'What can you NOW explain about the driving question that you could NOT explain in Lesson 1? Use the terms MA, VR, Efficiency, and Friction Heat in your answer.' Cards are posted on the DQB under the heading 'Lesson 7 — We Now Know.' The class reviews the DQB for any questions that remain unanswered — these are flagged for Lesson 8 (the culminating explanation lesson).	Point to the DQB: 'Look at the cards from Lesson 1. Read one of the 'We Wonder' questions from Week 1.' [Call on a learner to read one aloud — WAIT 8 seconds.] 'Can we answer that now? Thumbs up if yes, sideways if partial, down if still unsure.' Survey the room. Then say: 'Write your Lesson 7 update card. You must use all four terms: MA, VR, Efficiency, Friction Heat — in sentences, not a list. You have 3 minutes.' [WAIT 3 minutes of silent writing.] As learners post cards, read 2 aloud and say: 'This card says ___ — do others agree? Give a thumbs up or down.' Finally, circle any DQB questions still unanswered and say: 'These are your homework for tonight — bring one idea to Lesson 8 about how we could test or answer these.'	Driving Question Board Cumulative Update. Returning to the original driving question at the end of each lesson makes conceptual growth visible and metacognitive. Requiring all four key terms in the update card forces integration of today's new vocabulary (VR, Efficiency) with established concepts (MA, Friction Heat), reinforcing the coherent explanatory story of the unit.	Observable indicator: Each DQB card contains all four required terms used correctly in context. Teacher reads cards as they are posted — if fewer than 70% of cards use all four terms correctly, teacher addresses the gap at the start of Lesson 8 before moving to culminating explanation. A card that says 'energy is destroyed by friction' (rather than 'converted to heat') is used as a class correction moment — teacher reads it anonymously and asks: 'Agree or disagree — and why?'
Model Building Phase 📹 **VIDEO:** [Automotive mechanic: What I do and how much I make] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Angular Velocity (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Cumulative Class Model Update (10 min): The class returns to the large cumulative model diagram that has been built across Lessons 1–6 (posted on the wall or a dedicated section of the board). This model currently shows: force arrows, the lever/spanner system, MA calculation, and a heat arrow from the warm nut. Today's additions: (1) A double-headed arrow labelled 'dE (effort distance)' and 'dL (load distance)' with VR = dE/dL written beside the lever arm; (2) An efficiency equation box: Efficiency (%) = (MA/VR) × 100%; (3) A shaded 'energy loss' wedge between the input energy bar and the useful output energy bar, labelled 'Heat due to Friction —	Stand beside the cumulative class model: 'This is our class story of energy and machines — built over seven lessons. What is missing from it right now?' [WAIT 12 seconds. Cold-call 2 learners.] 'Exactly — we have MA but not VR or efficiency. Let's add them.' Invite a volunteer to draw the effort-distance and load-distance arrows onto the model diagram while the class gives directions. Then say: 'I need one more volunteer to draw the energy loss wedge. Where should it go — before or after the output? Why?' [WAIT 10 seconds.] After the model is updated, say: 'Copy these additions into your personal model notebook. Then write your one-sentence model	Cumulative Model Revision (Co-constructed). Physically adding to a shared visual model makes each lesson's conceptual contribution tangible and cumulative. The model now tells a complete causal story: long spanner → high VR → lower effort force (high MA) → BUT friction always present → efficiency < 100% → heat in nut. This directly answers both strands of the driving question and prepares learners for the Lesson 8 culminating explanation.	Observable indicator: The class model now contains all five elements: (1) force arrows with MA, (2) distance arrows with VR, (3) efficiency equation, (4) heat/friction wedge, (5) a direct visual link from the warm-nut phenomenon back to efficiency loss. Individual notebooks are checked for the model caption — accept any sentence that correctly uses VR, friction, heat, and efficiency < 100% together. Reject and re-prompt any caption that omits the causal link between friction and energy loss.

	always present, always > 0%.' Individual Model Refinement (5 min): Learners update their personal model in their notebooks, mirroring the class model additions. They write one 'model caption' sentence: 'The jua kali mechanic's long spanner has a higher VR because the effort travels farther, but friction in the threads always converts some work to heat, so efficiency is always below 100%.'	caption. 3 minutes, silent.' [WAIT 3 minutes.] Collect 3 notebooks at random to check caption quality — return with written feedback at the start of Lesson 8.		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners correctly distinguish between MA (force ratio) and VR (distance ratio) — or did they conflate the two? If more than a third of the class confused MA and VR, dedicate the first 10 minutes of Lesson 8 to a side-by-side comparison before the culminating explanation. (2) Were efficiency values calculated correctly? If systematic errors appeared (e.g., VR inverted, MA not retrieved accurately from Lesson 6), provide a corrected worked example at the start of Lesson 8. (3) Did learners independently connect the efficiency shortfall to the warm-nut phenomenon from Lesson 1, or did this connection require explicit teacher prompting? If prompting was needed, consider whether the DQB has been used consistently across all seven lessons — re-reading Lesson 1 DQB entries aloud at the start of Lesson 8 may help. (4) Were all groups able to set up and measure the lever system within 25 minutes? If not, identify which procedural steps caused the most delay and simplify the setup for Lesson 8's culminating activity. (5) Were learner DQB update cards of sufficient quality to use as formative data for Lesson 8 planning? Sort cards into three piles (fully answered the driving question, partially answered, still confused) and use the 'partially' and 'still confused' piles to design Lesson 8 opening moves.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners measured effort distance and load distance on a lever model at two different fulcrum positions, observed efficiency data for four different machines (lever, single pulley, wheelbarrow, screw jack) displayed in a gallery walk, and watched a video clip showing a pulley system at a posho mill near Eldoret to calculate VR and efficiency independently.
What did I learn?	Velocity Ratio (VR) = effort distance ÷ load distance; Efficiency (%) = (MA ÷ VR) × 100%; no real machine achieves 100% efficiency because friction always converts some work input into heat; machines redistribute force and distance but cannot create energy — the Law of Conservation of Energy is preserved and the warm wheel nut from Lesson 1 is the physical evidence of this energy conversion.
How does this explain the phenomenon?	The jua kali mechanic's long spanner extension increases the effort distance relative to the load distance, giving a high VR and high MA — so less muscular force is needed. However, friction in the wheel nut threads and spanner joints converts some of the work input into heat (the warm nut observed in Lesson 1), meaning the useful work output is always less than the total work input. This is why efficiency is always below 100% and why no machine can produce more energy than is put into it.

LESSON 8 (80 minutes): Synthesis, Model Finalisation, and DQB Resolution: The Full Story of the Jua Kali Mechanic	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate and synthesise all learning from the unit by constructing a complete, evidence-based explanation of the anchoring phenomenon — the jua kali mechanic loosening a wheel nut — integrating mechanical advantage, work, energy conservation, power, efficiency, and friction into a single coherent model.
Knowledge	Learners demonstrate that: (1) work is force multiplied by distance and is conserved across a simple machine ($W_{in} \approx W_{out} + W_{friction}$); (2) mechanical advantage is the ratio of load to effort and is greater than 1 when the effort arm is longer than the load arm; (3) energy is not created by a machine — it is transformed and, in real machines, partially dissipated as heat through friction; (4) efficiency is always less than 100% in real machines; and (5) power is the rate of doing work.
Skills	Learners are able to: annotate a force-and-energy diagram of a lever system; calculate mechanical advantage, velocity ratio, and efficiency for the spanner-lever system; present and justify a cumulative model using evidence gathered across all eight lessons; resolve questions on the Driving Question Board using specific lesson evidence; and co-construct a summary formula table.
Attitudes	Learners appreciate that scientific understanding is built incrementally through evidence, revision, and collaboration — mirroring the practical problem-solving ingenuity of Kenya's jua kali sector. They value intellectual honesty in acknowledging that no machine beats the law of conservation of energy.
Key Inquiry Question	Why can a jua kali mechanic in Nairobi loosen a wheel nut effortlessly with a long spanner when a short one fails — and where does the energy go when friction makes the nut hot?
Purpose in Storyline	This is the final lesson of the unit. Learners bring together every concept — work, energy, kinetic and potential energy, mechanical advantage, velocity ratio, efficiency, power, and friction — to construct the complete explanation of the anchoring phenomenon. They compare their Lesson 1 naive model with their final annotated model, resolve all DQB questions, and co-construct a unit summary table. The mechanic's long spanner, the warm wheel nut, and the apprentice's puzzlement are fully explained and closed.
Safety Notes	No laboratory equipment is used in this lesson. If printed model sheets are distributed, ensure scissors or sharp tools are not in use. Remind learners to handle any demonstration levers or spanners brought into class carefully, keeping them flat on desks when not in use.

B. LESSON OVERVIEW

Lesson 8 is the culminating synthesis lesson for Sub-Strand 1.4. Learners open by re-watching or re-reading the anchoring phenomenon (the Gikomba jua kali mechanic). They then move through five structured phases: (1) a Predict-Revisit in which students compare their Lesson 1 naive explanations with what they now know; (2) an Observe-Evidence Gallery in which groups rotate through posted evidence artefacts from Lessons 1–7 (calculations, graphs, diagrams, lab data); (3) a whole-class Final Explain session in which groups present annotated causal models integrating mechanical advantage, work-energy conservation, friction as a dissipative force, efficiency, and power; (4) a Driving Question Board resolution ceremony in which every sticky note is addressed and closed or redirected; and (5) a co-constructed summary table of all unit formulae anchored in the phenomenon context. Teacher uses cold-calling, structured gallery protocols, and 'press for mechanism' questioning throughout. The lesson ends with a teacher-facilitated exit ticket in which each learner writes a three-sentence scientific explanation of the phenomenon using at least four unit vocabulary terms.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Spout's reversing circuit and final assembly Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Mechanical Wave (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a printed copy of their own Lesson 1 prediction card (or see it projected). They read it silently for 2 minutes, then write in a different colour pen what they now know that their Lesson 1 self did not. In pairs, they share: 'The biggest thing I got wrong or incomplete in Lesson 1 was ___.' The class hears 4–5 pairs share aloud. The teacher re-plays the 60-second anchoring phenomenon video/image of the Gikomba mechanic using the pipe-extended spanner, and asks: 'Before we close this unit, use one sentence to describe what is happening using science — no maths yet, just mechanisms.' Learners write independently for 90 seconds. This activates retrieval of the full conceptual arc before group work begins.	Display the Lesson 1 phenomenon image alongside the question: 'What did you think was happening then? What do you know now?' Distribute or project Lesson 1 prediction cards. Say: 'Read your own Lesson 1 words. Do not change them yet — just notice what is missing.' Wait 2 minutes in silence. Then say: 'In your pairs, share the one biggest gap in your Lesson 1 thinking. You have 90 seconds — go.' After sharing, cold-call 4 pairs using the name sticks: 'Amina, what gap did you find in your Lesson 1 thinking?' After each response, push with: 'And what evidence from which lesson changed that thinking?' WAIT TIME: 12 seconds after each cold-call before accepting answers. Re-show the phenomenon and say: 'Now write one mechanism sentence — not a formula, a cause-and-effect sentence — explaining the mechanic's success.' Collect via show-of-hands for 3 volunteers to read aloud. Affirm scientific language use.	Retrieval + Contrastive Analysis. By explicitly comparing their Lesson 1 naive model with current knowledge, learners consolidate the conceptual journey of the unit. Research shows that noticing the gap between prior and current understanding strengthens long-term retention and reveals residual misconceptions for the teacher to address in Phase 3.	Observable indicator: Learners' one-sentence mechanism statements use at least two unit terms (e.g., 'effort arm', 'mechanical advantage', 'friction', 'energy dissipation', 'conservation', 'efficiency'). Teacher listens for residual misconceptions — e.g., 'the long spanner creates more energy' — and notes which learners to target in Phase 3 cold-calls. A learner who says 'the longer arm means less effort for the same turning effect, so energy is not created but redistributed' is at the target level.
Observe Phase 📹 **VIDEO:** Spout's reversing circuit and final assembly Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Mechanical Wave	Seven evidence stations are posted around the room, one per previous lesson. Each station has a large A3 sheet with: the key data, diagram, or graph from that lesson, and a prompt question connecting it to the phenomenon. Groups of 3–4 rotate every 4 minutes (timer displayed). At each station, one learner reads the artefact, one writes a sticky-note response to the prompt, and one records one 'lingering question' on	Set up stations before class. Say: 'Each station holds evidence your class collected or studied in a previous lesson. Your job is not to re-learn it — your job is to connect each piece of evidence back to the mechanic's problem. Read, discuss, and leave your sticky note answering the prompt.' Display a 4-minute countdown timer on the board. Circulate during rotations. At Station 1, prompt: 'How does this calculation show why the short	Evidence Gallery Walk with Claim-Evidence Linking. This strategy makes all seven lessons' worth of evidence simultaneously visible and forces learners to construct cross-lesson connections rather than treating each lesson as isolated content. The sticky-note protocol makes thinking visible and gives the teacher real-time formative data.	Observable indicator: Sticky notes at each station contain a phenomenon-linked claim with a reference to specific data (e.g., 'At Station 1, W_{in} = 120 J both ways, so the machine does not add energy — it only changes the force and distance'). Groups that write only vague generalities ('energy is conserved') without referencing numbers or diagrams are identified for targeted support in Phase 3. Teacher collects all sticky notes

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the DQB panel. Station contents: Station 1 — Work calculation comparing short vs long spanner (W = F × d); Station 2 — Lever diagram with effort arm and load arm labelled, MA = Load/Effort; Station 3 — Energy transformation flow diagram (chemical → kinetic → mechanical → heat); Station 4 — Efficiency calculation showing η < 100% for the spanner system; Station 5 — Power comparison (same work, faster vs slower mechanic); Station 6 — Friction as a force: graph of temperature vs number of failed attempts; Station 7 — LabXchange/Khan Academy screenshot of a pulley system showing VR vs MA. Groups add their sticky note to the station before rotating.	spanner failed?' At Station 3, ask: 'Where does the heat energy in the nut come from in this diagram?' At Station 4, ask: 'If efficiency is always less than 100%, can the mechanic ever get more work out than he puts in?' WAIT TIME: 10 seconds after each probe question. After all rotations, bring the class back: 'Which station gave your group the clearest answer to our driving question? Cold-call 3 groups: Ochieng's group — Station?' Affirm connections; challenge vague answers: 'You said the energy goes to heat — show me which arrow on the Station 3 diagram proves that.'		after the gallery for review.
Explain Phase 📹 **VIDEO:** Spout's reversing circuit and final assembly Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Mechanical Wave (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group (4–5 groups of 6 learners) presents their final annotated model of the jua kali phenomenon — a poster or digital slide showing: (a) a labelled lever diagram with effort arm (long spanner + pipe), load arm (nut radius), effort force, load force, MA calculation, and VR; (b) an energy flow diagram showing chemical energy (mechanic's muscles) → kinetic energy (arm movement) → mechanical work (on nut) + heat energy (friction); (c) an efficiency statement ('Because of friction, only ___% of the input work tightens the nut; the rest heats the metal'); (d) a power statement ('A stronger or faster mechanic does the same work in less time, increasing power output'). Each group has 3 minutes to present; the class has 2 minutes of structured peer questioning. After all groups, the teacher leads a 10-minute whole-class synthesis:	Introduce the presentation protocol: 'Each group will present for exactly 3 minutes. You must address three questions: (1) How does the long spanner create mechanical advantage? (2) Where does the energy go? (3) Why can't the mechanic get more work out than he puts in? After each group, I will call on two learners from other groups — not the presenter group — to ask a peer question or add evidence.' Cold-call peer questioners by name: 'Wanjiru, you are on questioning duty for Group 2. Prepare one question during their presentation.' WAIT TIME: 12 seconds after asking 'What is the mechanism?' before accepting an answer. During synthesis, write the class sentence on the board in stages, filling in blanks: 'The long spanner increases _______ by increasing the _______ arm, so the mechanic applies less _______ over a	Structured Academic Controversy / Press-for-Mechanism Presentation Protocol. Requiring groups to explain mechanism (not just state facts) and exposing them to peer questioning forces deep elaboration. The co-constructed synthesis sentence is a whole-class sensemaking anchor that integrates all unit concepts into one causal narrative tied to the phenomenon.	Observable indicator: Final models correctly show: MA > 1 for the long spanner (e.g., MA = 3.5); W_in > W_out with the difference accounted for by heat; efficiency < 100%; and power as work/time. The synthesis sentence on the board is evaluated by the teacher for causal completeness. Learners who cannot distinguish between 'less force needed' and 'less work done' are identified for individual follow-up. Teacher notes whether the energy flow diagram shows heat as a product of friction — not a mysterious disappearance.

	'What is the one sentence that ties mechanical advantage, energy conservation, and friction together for the mechanic's problem?' The class co-writes this sentence on the board.	greater _______ to do the same _______. Because friction dissipates energy as _______, the nut gets _______ and efficiency is always less than _______%. No machine can create _______.' Correct residual misconceptions explicitly: 'I heard someone say the long spanner makes more energy — let's test that with numbers from Station 1.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Spout's reversing circuit and final assembly] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Mechanical Wave (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	The full DQB from all eight lessons is displayed at the front. Each sticky note question is read aloud by the learner who posted it (or by the teacher if the author is unknown). The class votes — thumbs up, sideways, or down — on whether the question is: (A) fully answered by our unit, (B) partially answered, or (C) beyond our current unit (redirected to a future topic). For Category A questions, a volunteer states the answer in one sentence using unit vocabulary. For Category B, the teacher acknowledges the partial explanation and names the gap. For Category C, the teacher writes 'To Strand 2.0 / 3.0 / Future Learning' next to the question and says: 'This is how science works — one answered question opens new ones.' The mechanic's original two mysteries — mechanical advantage and heat — are given a formal 'resolution moment': the teacher reads the original driving question aloud, and three learners each answer one part of it in sequence.	Stand at the DQB and say: 'This board holds every question we've asked about the mechanic's problem since Lesson 1. Today we close it — or we redirect it. I'll read each question. You vote with your thumbs.' Model the three categories with one example each before starting. For Category A resolutions, cold-call learners who posted the original question: 'Kipchoge, you asked this in Lesson 2 — do you have the answer now?' WAIT TIME: 10 seconds. For unresolved questions, validate intellectual honesty: 'It is a sign of a good scientist to know what they don't yet know.' For the driving question resolution, say: 'The question that started this whole unit was: Why can a jua kali mechanic loosen a wheel nut effortlessly with a long spanner when a short one fails — and where does the energy go when friction makes the nut hot? Three of you will each answer one part. Zawadi — the spanner. Otieno — the energy. Amara — the friction.' Give each 30 seconds. Applaud the closure. Physically remove answered sticky notes and place them in a 'Resolved' envelope.	Driving Question Board Resolution (Metacognitive Closure Protocol). Formally resolving each question makes the arc of scientific inquiry visible and gives learners ownership of their learning journey. Redirecting unanswered questions to future strands models authentic scientific practice — science is cumulative and open-ended. The 'resolution moment' for the anchoring phenomenon creates an emotionally satisfying narrative closure tied to the real-world Kenyan context.	Observable indicator: At least 80% of DQB questions are categorised by learner consensus (not just teacher direction). Learner answers for Category A questions include unit-specific vocabulary and a mechanism (not just a formula). Teacher listens for whether learners can articulate why energy conservation means MA > 1 does not imply free energy — this is the key conceptual target of the entire unit.
Model	The teacher facilitates a whole-	Draw the empty summary table on	Co-Constructed Summary Table	Observable indicator: Exit tickets

Building Phase 📹 **VIDEO:** Spout's reversing circuit and final assembly Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Mechanical Wave (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	class co-construction of a summary table on the whiteboard (learners copy into their notebooks). The table has five columns: Concept | Formula | Unit | What it means for the mechanic | Kenya example. Learners contribute each row by recalling from memory before the teacher fills in any blanks. Rows include: Work (W = Fd, Joules); Kinetic Energy (KE = ½mv², Joules); Gravitational PE (GPE = mgh, Joules); Mechanical Advantage (MA = Load/Effort, no unit); Velocity Ratio (VR = Effort distance/Load distance, no unit); Efficiency (η = (MA/VR) × 100%, %); Power (P = W/t, Watts). For each row, one learner provides the 'What it means for the mechanic' cell and one provides a Kenya example beyond the spanner (bicycle gears in Kisumu, water pump pulley in Kericho, wheelbarrow on a Rift Valley farm, maize-grinding mill in Eldoret). The lesson closes with a 5-minute individual EXIT TICKET: learners write three sentences explaining the phenomenon using at least four vocabulary terms from the summary table.	the board. Say: 'You are going to fill this in from memory — no notes yet. I'll ask for volunteers row by row, but I reserve the right to cold-call.' For each row, ask: 'Who can give me the formula?' then 'Who can tell me what this means for the mechanic?' then 'Who has a Kenya example that is NOT the spanner?' Cold-call 2 learners per row to ensure broad participation. WAIT TIME: 10 seconds before accepting volunteered answers. When a learner gives an incorrect formula, say: 'Interesting — let's test that against our Station 1 numbers. Does it give us the right answer?' Guide correction through evidence, not just assertion. For the Kenya examples, probe: 'The bicycle gear in Kisumu — is the MA greater than 1 or less than 1 in high gear? Why does that matter for a cyclist climbing the Rift Valley escarpment?' For the exit ticket, say: 'Three sentences. Four vocabulary terms minimum. Go — you have five minutes.' Collect exit tickets at the door. Preview: 'In Strand 2.0 we will meet waves — and you will find that energy conservation follows us there too.'	(Generative Retrieval Practice). Having learners recall and contribute each row — rather than receiving a completed table — forces generative processing, which research shows significantly increases retention compared to passive copying. Anchoring every formula to the phenomenon and to diverse Kenyan contexts builds transferable understanding rather than isolated symbol manipulation.	contain three sentences that: (1) name a specific mechanism (e.g., 'the longer effort arm increases mechanical advantage'), (2) reference energy conservation or efficiency (e.g., 'energy input equals useful work plus heat from friction'), and (3) use at least four of the following terms correctly: work, energy, force, distance, mechanical advantage, efficiency, power, friction, velocity ratio, conservation. Teacher sorts exit tickets into three piles (Full explanation / Partial / Needs follow-up) and uses these to plan any necessary consolidation at the start of the next strand.

D. TEACHER REFLECTION

After delivering Lesson 8, reflect on the following questions before moving to Strand 2.0:

1. CONCEPT CLOSURE — Did the exit tickets show that learners can articulate both halves of the driving question (mechanical advantage AND energy dissipation)? What proportion reached the full-explanation threshold? Were there systematic gaps (e.g., most learners explaining the spanner but not the heat, or vice versa)?

2. DQB QUALITY — Were there questions on the DQB that were never adequately resolved? If so, identify whether these represent curriculum gaps, instructional gaps, or genuinely advanced questions to be revisited in later strands. Note these for your strand transition.

3. MODEL COMPARISON — When learners compared their Lesson 1 naive models with their final models, what were the most common conceptual shifts? What was the most persistent misconception across the unit (e.g., 'machines create energy', 'efficiency can exceed 100%', 'friction is just a nuisance and not an energy transformation')? How will you address this in Strand 2.0 contexts?

4. DIFFERENTIATION — Were all learner groups able to present a complete annotated model, or did some groups produce only partial models? Plan targeted consolidation for those groups before introducing Waves and Optics, as energy conservation underpins that strand as well.

5. KENYA CONTEXT RICHNESS — Did learners generate their own Kenyan examples in the summary table (beyond the spanner and the ones you suggested)? The depth and creativity of their examples is a proxy for genuine conceptual transfer. If learners defaulted to textbook examples from other countries, plan to use more local phenomena as anchors in Strand 2.0.

6. TIMING — An 80-minute lesson with five phases is ambitious. Note which phase overran and adjust the Phase weighting for future unit closures. If Phase 3 (model presentations) consistently runs long, consider limiting each group to 2 minutes with a strict timer visible to learners.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that a jua kali mechanic in Gikomba, Nairobi, failed to loosen a wheel nut with a short spanner but succeeded effortlessly when he extended the spanner arm with a metal pipe. We also observed that the wheel nut became noticeably warm after repeated failed attempts with the short spanner, even though the nut did not move.
What did I learn?	We learned that: (1) a longer effort arm increases mechanical advantage (MA = Load ÷ Effort), allowing a smaller force to produce a larger turning effect on the nut; (2) the total work input equals useful work output plus work done against friction ($W_{in} = W_{out} + W_{friction}$), so the machine does not create energy — it redistributes force and distance; (3) friction converts mechanical energy into thermal energy, which is why the nut gets hot; (4) efficiency (η = MA/VR × 100%) is always less than 100% in real machines because of friction; and (5) power (P = W/t) measures how quickly the mechanic does work — a stronger or faster mechanic has greater power output.
How does this explain the phenomenon?	We can now fully explain the anchoring phenomenon: the long spanner creates a greater mechanical advantage by increasing the effort arm length, meaning the mechanic applies a smaller force over a larger distance to produce the same work (turning the nut). The machine does not create extra energy — it trades force for distance. The heat in the nut is explained by the work-energy principle: work done against friction is transformed into thermal energy, which dissipates into the metal. Because of this friction loss, the mechanic must always put in more energy than he gets out as useful work — no machine can exceed 100% efficiency, and energy is always conserved across the whole system.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.